

M5 agentic experiment report — PARTIAL

Snapshot generated 2026-10-07T21:25:03+00:00 (UTC). This report uses current results under this checkout; repository examples and dissertation narrative are not treated as outputs of this execution.

Overall execution status: PARTIAL. There are 565 completed current-scope configured runs and 30 recorded LLM calls in those completed runs. A completed deterministic control is not an LLM agent experiment. This is the latest available report; a PARTIAL or BLOCKED report is not a completed experiment.

Execution status and remaining work

COMPLETE means every configured policy/scenario/seed/origin run has a per-run summary and the configured number of daily rows. PARTIAL means execution has started or some runs are complete. BLOCKED records a stated missing prerequisite. UNRUN means no completed run or startup evidence is available. Completion counts are calculated from artifacts, even when pipeline status is stale; completion does not by itself establish audit validity, statistical power, or model quality. Overall execution completion applies to stages marked required; superseded or archived studies retain their actual artifacts separately. Configured-run completion also does not satisfy unimplemented dissertation requirements.

Stage	Status	Required	Complete / target	Decision days	Policies	Carrier
m5_smoke	COMPLETE	yes	12 / 12	4	B1, D0, D1	template
ml_m5_smoke	COMPLETE	yes	16 / 16	4	B1, B2, B3, B4	template
baseline_pilot	COMPLETE	yes	16 / 16	7	B1, B2, B3, B4	template
main_study_stage1	PARTIAL	no	20 / 480	28	B1, B2, B3, B4	template
foundation_30series	BLOCKED	yes	0 / 360	28	B1, B5	template
llm_architectures_ollama	BLOCKED	no	0 / 18	4	B4, B6, B7, B8, B9, B10	template
llm_prose_reference_30series	PARTIAL	no	1 / 360	28	B3, B4	template
llm_prose_study_ollama	BLOCKED	yes	0 / 540	28	B4, B9, B10	prose
llm_prose_study_reference	COMPLETE	no	120 / 120	4	B3, B4	template
llm_template_architectures_30series	BLOCKED	yes	0 / 900	28	B6, B7, B8, B9, B10	template
parallel_runner_validation	COMPLETE	no	8 / 8	1	B1, B2, B3, B4	template
six_hour_agentic_prose	COMPLETE	yes	8 / 8	2	B3, B4, B9, B10	prose
six_hour_numeric	COMPLETE	yes	96 / 96	7	B1, B2, B3, B4	template
six_hour_numeric_extension	COMPLETE	yes	264 / 264	7	B1, B2, B3, B4	template
six_hour_structured_controls	COMPLETE	yes	4 / 4	2	B3, B4	template

main_study_stage1: pipeline-reported status=paused; observed artifact status=PARTIAL. In-progress run directories without complete summaries: 3.

foundation_30series Required stage prerequisite/blocker: Chronos metadata pinned and libraries validated; remaining weight file requires us.aws.cdn.hf.co network route

foundation_30series readiness observation: Pinned Chronos metadata revision has been obtained; it does not attest downloaded model weights or tested foundation inference.

foundation_30series: pipeline-reported status=blocked; observed artifact status=BLOCKED. In-progress run directories without complete summaries: 0.

llm_architectures_ollama Auxiliary stage's historically reported prerequisite: Official model registered; awaiting live development grounding and throughput/storage preflight

llm_architectures_ollama readiness observation: Verified registry download and local registration are attested by ollama_model_identity.json; this alone does not validate functional inference or full-study readiness.

llm_architectures_ollama readiness observation: Configured model digest matches registered identity.

llm_architectures_ollama: pipeline-reported status=blocked; observed artifact status=BLOCKED. In-progress run directories without complete summaries: 0.

llm_prose_reference_30series: pipeline-reported status=paused; observed artifact status=PARTIAL. In-progress run directories without complete summaries: 1.

llm_prose_study_ollama Required stage prerequisite/blocker: Official model registered; awaiting live development grounding and throughput/storage preflight

llm_prose_study_ollama readiness observation: Verified registry download and local registration are attested by ollama_model_identity.json; this alone does not validate functional inference or full-study readiness.

llm_prose_study_ollama readiness observation: Configured model digest matches registered identity.

llm_prose_study_ollama: pipeline-reported status=blocked; observed artifact status=BLOCKED. In-progress run directories without complete summaries: 0.

llm_prose_study_reference: pipeline-reported status=complete; observed artifact status=COMPLETE. In-progress run directories without complete summaries: 0.

llm_template_architectures_30series Required stage prerequisite/blocker: Official model registered; awaiting live development grounding and throughput/storage preflight

llm_template_architectures_30series readiness observation: Verified registry download and local registration are attested by ollama_model_identity.json; this alone does not validate functional inference or full-study readiness.

llm_template_architectures_30series readiness observation: Configured model digest matches registered identity.

llm_template_architectures_30series: pipeline-reported status=planned; observed artifact status=BLOCKED. In-progress run directories without complete summaries: 0.

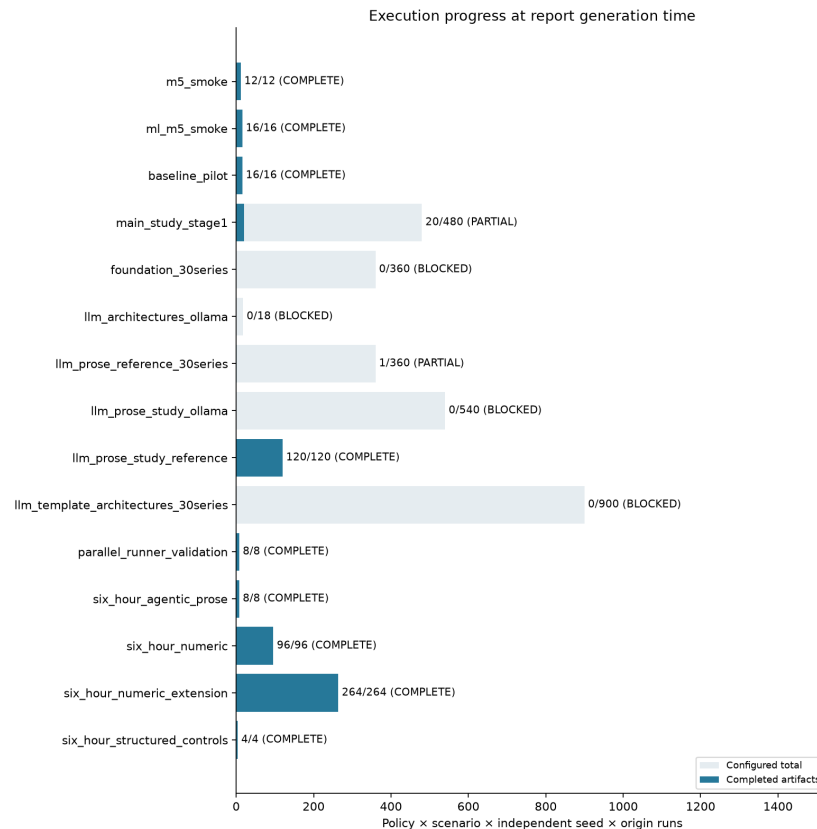
parallel_runner_validation: pipeline-reported status=complete; observed artifact status=COMPLETE. In-progress run directories without complete summaries: 0.

six_hour_agentic_prose: pipeline-reported status=complete; observed artifact status=COMPLETE. In-progress run directories without complete summaries: 0.

six_hour_numeric: pipeline-reported status=complete; observed artifact status=COMPLETE. In-progress run directories without complete summaries: 0.

six_hour_numeric_extension: pipeline-reported status=complete; observed artifact status=COMPLETE. In-progress run directories without complete summaries: 0.

six_hour_structured_controls: pipeline-reported status=complete; observed artifact status=COMPLETE. In-progress run directories without complete summaries: 0.



Configured run completion; worker and merged copies of the same run are counted once.

Model preparation and functional readiness

Recorded local model: ega-llama3.2:3b-cloud30; registered digest: bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d; parameter size: 3.2B; quantization: Q4_K_M. Source attestation: official Ollama registry verified SHA256 content; locally derived Modelfile cloud30. This establishes the recorded verified download/registration step, not successful end-to-end experimental decisions.

Readiness records below are separate from completed simulation results: model registration, validated structured responses, measured throughput, configuration freeze and a completed study are different milestones. Historical network failures in earlier audit/validation snapshots do not override a later verified identity. Successful preflight calls do not count as completed full-study runs.

recorded preparation source: results/local_model_development_30series_batch1/development_manifest.json

Readiness summary: {"complete": true, "day": 1700, "development_only": true, "model_revision": "bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d", "pilot_runs": 0, "series": 30}

recorded preparation source: results/logs/ollama_cloud30_functional_readiness.json

Readiness summary: {"elapsed_seconds": 1.1468176409998705, "model_revision": "bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d"}

Recorded functional check: requested structured value matched=True; positive actual input/output usage=True; prompt tokens=36; completion tokens=8; elapsed seconds=1.147. This validates this small structured call, not supplier-rule accuracy, inventory performance or full-study throughput.

recorded preparation source: results/logs/six_hour_preflight.json

This readiness snapshot records agentic configs awaiting freeze/registration: /workspace/MasterDissertation/results/run_configs/six_hour_agentic_prose.yaml,

/workspace/MasterDissertation/results/run_configs/six_hour_agentic_template.yaml. Registered stage configs and actual run artifacts take precedence if this preparation snapshot later becomes stale.

recorded preparation source: results/run_configs/local_model_resource_preflight.json

Readiness summary: {"model_revision":
"bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d"}

Measured resource preflight: passed=False; blockers=['Projected artifact storage plus reserve exceeds current free disk']. Measurements: {"max_actual_zip_deflate_sample_ratio": 0.15718854449707842, "mean_call_seconds": 92.26134998425005, "mean_request_artifact_bytes": 13287.5, "p95_call_seconds": 123.14005513200027, "real_call_records": 4}. Successful-path model-only projection at mean=6,006.214 days, at p95=8,016.418 days. These are development-sample planning estimates, not elapsed full-experiment results; they exclude additional training, optimization, scheduling, retries and reports.

recorded preparation source: results/run_configs/local_model_resource_preflight_incremental.json

Incremental resource evidence: 2 actual development responses; first extraction request=243.206 seconds; planned successful-path requests before retries=438,480. The recorded same-latency multiplication is 1,234.271 days. This is a scale illustration from an initial request, not a representative completion forecast or an executed study duration.

Incremental estimate limit: A single extraction request is not a representative latency estimate for all roles, memory lengths, cache effects, validation failures and retries.

Incremental estimate limit: The coarse same-latency multiplication is a scale illustration, not a completion forecast.

Incremental estimate limit: Full inference cannot be called feasible based on an installed model or a trivial response.

Incremental estimate limit: No full-study LLM evaluation has started; CPU/GPU/provider choice and common serving batch remain development decisions.

recorded preparation source: results/run_configs/six_hour_runtime_preflight.json

This readiness snapshot records agentic configs awaiting freeze/registration:
/workspace/MasterDissertation/results/run_configs/six_hour_agentic_template.yaml. Registered stage configs and actual run artifacts take precedence if this preparation snapshot later becomes stale.

recorded preparation source: results/run_configs/six_hour_supervisor_status.json

Readiness summary: {"phase": "all_supplied_studies_complete", "updated_at_utc":
"2026-10-07T20:56:49.458868+00:00"}

This readiness snapshot records agentic configs awaiting freeze/registration:
/workspace/MasterDissertation/results/run_configs/six_hour_agentic_template.yaml. Registered stage configs and actual run artifacts take precedence if this preparation snapshot later becomes stale.

recorded preparation source: results/run_configs/supervisor_status.json

Readiness summary: {"blocker": "Projected artifact storage plus reserve exceeds current free disk; detailed truthful estimates saved in local_model_resource_preflight.json", "functional_usage": {"completion_tokens": 8, "prompt_tokens": 36, "prompt_tokens_details": {"cached_tokens": 35}, "total_tokens": 44}, "model_name": "ega-llama3.2:3b-cloud30", "model_revision":
"bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d", "phase": "blocked", "preflight_passed": false, "updated_at_utc": "2026-10-07T20:07:08.033226+00:00"}

Recorded foundation metadata: amazon/chronos-t5-tiny at revision
29d808298f1a62493e7b9a5e08529d0d930fa189. Metadata retrieval is not evidence that the model.safetensors download or inference succeeded.

Controlled grounding benchmarks — actual current artifacts

Incremental and final snapshots from the same development folder can describe the same responses. They are shown as recorded snapshots and are not pooled or counted as independent replications. Retained development configurations with different batch sizes also remain separate from frozen evaluation studies.

Source: results/grounding_templates/grounding_results.json; evidence role=recorded grounding; reader mode=deterministic_template; model=not recorded; model revision=not recorded; development_only=False; incremental=False; complete=not recorded; total LLM calls=0; errors=0; tokens=0. Controlled carrier benchmark, not a validated natural-language dataset. No expected labels are sent to the model.

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
normal / deterministic_template	True	1.000	1.000	0 / 0	0	False
aggregate_moq / deterministic_template	True	1.000	1.000	0 / 0	0	False
foreign_unit_moq / deterministic_template	True	1.000	1.000	0 / 0	0	False
pack_change / deterministic_template	True	1.000	1.000	0 / 0	0	False
capacity_cut / deterministic_template	True	1.000	1.000	0 / 0	0	False
eligibility_change / deterministic_template	True	1.000	1.000	0 / 0	0	False

Source: results/local_model_development_30series_batch1/grounding/grounding_results.json; evidence role=recorded grounding; reader mode=local LLM development sample; model=not recorded; model revision=bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d; development_only=True; incremental=False; complete=not recorded; total LLM calls=4; errors=0; tokens=11690. Field-complete controlled carriers; this sample is not a natural-language production benchmark.

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
development batch / template	False	—	—	1 / 1	—	—
development batch / template	False	—	—	1 / 1	—	—
development batch / prose	False	—	—	1 / 1	—	—
development batch / prose	False	—	—	1 / 1	—	—

Source: results/local_model_development_30series_batch16/grounding/grounding_results.json; evidence role=recorded grounding; reader mode=local LLM development sample; model=not recorded; model revision=bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d; development_only=True; incremental=False; complete=not recorded; total LLM calls=2; errors=0; tokens=9659. Field-complete controlled carriers; this sample is not a natural-language production benchmark.

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
development batch / template	False	—	—	0 / 14	—	—

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
development batch / prose	False	—	—	1 / 15	—	—

Source: results/local_model_development_30series_batch16/grounding/incremental_grounding_results.json; evidence role=recorded grounding; reader mode=local LLM development sample; model=not recorded; model revision=bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d; development_only=True; incremental=True; complete=False; total LLM calls=not recorded; errors=not recorded; tokens=not recorded. Labels are compared by evaluator only; they were not provided to the model. Schema validity does not establish grounding correctness.

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
development batch / template	False	1.000	0.067	0 / 14	—	—
development batch / prose	False	0.000	0.000	1 / 15	—	—

Carrier	Source docs	Constraints returned	Schema valid / finish	Input / output tokens	Seconds
template	15	1	True / stop	4812 / 152	243.206
prose	15	1	True / stop	4588 / 107	201.623

Schema-valid responses can still omit or misground rules. Missing residual-error/escalation fields remain unmeasured rather than zero. Development samples do not establish a favorable LLM effect or confirmatory full-study performance.

The measured development samples above contain actual extraction failures: whole-set mismatches, omissions or false constraints remain errors despite valid JSON and a normal finish reason.

Source: results/six_hour_development_grounding/grounding_results.json; evidence role=recorded grounding; reader mode=real_llm; model=ega-qwen2.5:0.5b-sixhour; model revision=21539788ff12776c3460262e890e2d77f98d99e5ce24a1726a9582a31be23e44; development_only=True; incremental=False; complete=True; total LLM calls=4; errors=4; tokens=15686. Four controlled development cases with all 30 entities; not evaluation or general natural-language accuracy.

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
development batch / template	False	—	—	— / —	—	—
development batch / template	False	—	—	— / —	—	—
development batch / prose	False	—	—	— / —	—	—
development batch / prose	False	—	—	— / —	—	—

Case / carrier / batch	Exact / false / omitted rules	Calls	Seconds	Request result
1 / template / 1	0 / unmeasured / 1	1	35.684	request failed
2 / template / 1	0 / unmeasured / 1	1	36.045	request failed
3 / prose / 1	0 / unmeasured / 1	1	32.902	request failed
4 / prose / 1	0 / unmeasured / 1	1	31.013	request failed

Development case 1 request error: LLM request failed after 1 attempts: Truncated model output; increase max_tokens or reduce batch size

Development case 2 request error: LLM request failed after 1 attempts: Truncated model output; increase max_tokens or reduce batch size

Development case 3 request error: LLM request failed after 1 attempts: Truncated model output; increase max_tokens or reduce batch size

Development case 4 request error: LLM request failed after 1 attempts: Truncated model output; increase max_tokens or reduce batch size

These exact-rule counts and durations come from the recorded development cases. Failed requests can have recorded omissions without a validated returned-rule set; missing false-rule/precision fields remain unmeasured. Serving-model selection uses these development samples, which are excluded from inventory-policy comparisons.

Source: results/six_hour_qwen15_development/grounding_results.json; evidence role=recorded grounding; reader mode=real_llm; model=ega-qwen2.5:1.5b-sixhour; model revision=14bdd46019879e18cd27f21b46e2c377bf801e827e2abb46be2daedcedaf0f64; development_only=True; incremental=False; complete=True; total LLM calls=4; errors=0; tokens=14152. Four controlled development cases; serving batch selection only. Expected labels are excluded from all requests.

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
development batch / template	False	—	—	— / —	—	—
development batch / prose	False	—	—	— / —	—	—
development batch / template	False	—	—	— / —	—	—
development batch / prose	False	—	—	— / —	—	—

Case / carrier / batch	Exact / false / omitted rules	Calls	Seconds	Request result
1 / template / 1	0 / 1 / 1	1	34.768	response received; see exact-rule counts
2 / prose / 1	0 / 1 / 1	1	30.906	response received; see exact-rule counts
3 / template / 4	0 / 4 / 4	1	55.303	response received; see exact-rule counts
4 / prose / 4	3 / 1 / 1	1	52.360	response received; see exact-rule counts

These exact-rule counts and durations come from the recorded development cases. Failed requests can have recorded omissions without a validated returned-rule set; missing false-rule/precision fields remain unmeasured. Serving-model selection uses these development samples, which are excluded from inventory-policy comparisons.

Source: results/six_hour_qwen15_documented_format_development/grounding_results.json; evidence role=recorded grounding; reader mode=real_llm; model=ega-qwen2.5:1.5b-sixhour; model revision=14bdd46019879e18cd27f21b46e2c377bf801e827e2abb46be2daedcedaf0f64; development_only=True; incremental=False; complete=True; total LLM calls=2; errors=0; tokens=8655. Four controlled development cases; serving batch selection only. Expected labels are excluded from all requests.

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
development batch / template	False	—	—	— / —	—	—
development batch / prose	False	—	—	— / —	—	—

Case / carrier / batch	Exact / false / omitted rules	Calls	Seconds	Request result
1 / template / 4	0 / 4 / 4	1	63.954	response received; see exact-rule counts
2 / prose / 4	0 / 4 / 4	1	57.787	response received; see exact-rule counts

These exact-rule counts and durations come from the recorded development cases. Failed requests can have recorded omissions without a validated returned-rule set; missing false-rule/precision fields remain unmeasured. Serving-model selection uses these development samples, which are excluded from inventory-policy comparisons.

Deterministic-template results measure the parser control. Controlled LLM corpus results measure that recorded carrier/case set; neither is a validated natural-language supplier population or evidence that pending M5 studies finished.

Data, design and interpretation limits

M5 supplies historical observed sales, calendar and prices. Sales are used as an exogenous demand proxy and may be censored by historical stockouts; latent demand is not recovered. Inventory, replenishment, supplier documents, lead times, capacity, approval, faults and operational constraints are simulated. Simulated run costs and harm outcomes are not measured retailer operations.

The prepared data are selected panels, not all 30,490 bottom-level M5 series or the official competition evaluation. Fault rates are synthetic and uncalibrated unless a configuration explicitly records calibration. Simulated delayed oracle approval is a reviewer model, not a human audit study. Findings are conditional on this panel, window, solver, forecast training, fault definitions and approval model.

Prepared panel	Serie s	Historica l days	Selection	Syntheti c sales?
data/processed/m5	30	1941	{"items": ["FOODS_1_001", "FOODS_2_001", "FOODS_3_001"], "stores": "all"}	False
data/processed/m5_it em1	10	1941	{"items": ["FOODS_1_001"], "stores": "all"}	False
data/processed/m5_s moke	2	1941	{"items": ["FOODS_1_001"], "stores": ["CA_1", "CA_2"]}	False

The main stage-1 design requires 480 runs: four deterministic policies × four fault scenarios × thirty independent seeds × one origin. It uses 28 decision days per run. The shortened study has its own registered seed count and horizon. Other stages retain their own panel, carrier and horizon; their cost totals must not be pooled as interchangeable replications. The remaining scenarios and any policies absent from registered configs are unrun in this report.

Policy	Actual architecture
B1	Deterministic seasonal-naive forecast + order-up-to control
B10	Typed LLM roles + critic + per-decision evidence gate
B2	Trained LightGBM quantile forecast + (s,S) control

Policy	Actual architecture
B3	Trained native GRU/negative-binomial forecast + stochastic MILP
B4	B3 + deterministic evidence gate; no LLM
B5	Chronos forecast adapter; requires separate weights
B6	Single generalist LLM + shared numerical tools
B7	LLM roles with free-form shared messages + validated action boundary
B8	Typed LLM roles without critic
B9	Typed LLM roles + critic, fixed autonomy
D0	Deterministic seasonal-forecast optimizer software control; no LLM
D1	D0 + deterministic evidence gate software control; no LLM

The native deep model is a trained GRU with a negative-binomial likelihood, not an exact DeepAR or TFT reproduction. LLM roles share deterministic numerical tools and schema validation. LLM calls, errors and explicit fallback decisions are reported separately; a held decision or deterministic fallback does not prove successful semantic reasoning. A seasonal forecast override changes the baseline and must be read in that stage's config.

Alignment with the user's revised dissertation

Reference document: [dissertation_revised_business_school.html](#). Uploaded-file SHA-256: 8ae699667fa02f932f4c2584af0a3895021dc925e3119412bb06b238dc87de1c. Extracted-text SHA-256: fcd4e9939c99143dac478947209a6fcdac759ec3f0b8534d9d155b417511fd91. The document supplies the research specification and expected managerial discussion; its expected findings are not results of this execution.

The user requested the complete experiment and accepted all thirty prepared item-store series for the B9/B10 agent comparison and matched references. A previous ten-series reference or two-series connectivity pilot is a distinct panel, not evidence that the thirty-series comparison has finished. Each stage below must retain its own actual dataset manifest, matched seeds, origins, decision horizon and model digest. This thirty-series panel is still a subset of full M5.

Exact revised research questions and evidence coverage

RQ1 — Performance. How does the proposed system compare with classical and forecast-plus-optimizer baselines on total cost, service, stockouts, and bullwhip amplification when the data state is clean?

Current evidence scope — PARTIAL: Current summaries and forecast holdouts support descriptive cost/service, stockout and bullwhip measurement for the completed controls. Full matched agent-versus-baseline performance remains conditional on completed stage grids.

Implementation audit's required contrast: Clean-state cost/service/stockouts/bullwhip across classical, forecast+optimizer, and agentic policies

Implementation audit snapshot status: partial until foundation weights and real model run. This describes implementation/design coverage at audit time, not an empirical hypothesis verdict; live execution is reported separately.

Remaining implementation or design gap: No LLM-only comparator, no TFT/Moirai, subset-aware forecast scores only.

RQ2 — Evidence quality. When the observed state is degraded, does evidence-gated autonomy reduce harmful executed orders relative to systems that continue operating with fixed autonomy?

Current evidence scope — PARTIAL: Harm, true-constraint violation, reference-deviation and hold counts are available for completed simulations. They do not yet establish an agent gate effect over the complete revised design; direct B10-B9 analysis waits for a full matched grid.

Implementation audit's required contrast: Fixed vs gated autonomy with corrupted observations

Implementation audit snapshot status: implementable gate effect; semantic component partial. This describes implementation/design coverage at audit time, not an empirical hypothesis verdict; live execution is reported separately.

Remaining implementation or design gap: LLM triage records hypotheses but cannot repair state or clear hard failures. Current checks catch the programmed failures; no implemented ambiguous-reconciliation intervention demonstrates semantic repair benefit. Reference deviations and true violations must be reported separately.

RQ3 — Constraint grounding. How accurately can the system translate realistic replenishment constraints—such as pack sizes, supplier minimums, unit conversions, and assortment eligibility—into typed, verifiable rules?

Current evidence scope — PARTIAL: Live extraction and controlled grounding evaluations are required for exact typed-rule accuracy and errors reaching the solver. A model call count alone is not a grounding-accuracy result; template parser controls do not measure an LLM's semantic contribution.

Implementation audit's required contrast: Typed grounding of local/coupled/unit-conversion/validity rules

Implementation audit snapshot status: controlled grounding implementable. This describes implementation/design coverage at audit time, not an empirical hypothesis verdict; live execution is reported separately.

Remaining implementation or design gap: Supplied six cases are explicit synthetic rules, not a realistic supplier/workbook or multilingual benchmark. Generic XLSX semantics and ambiguous external-document corpus are absent. Prose verification can accept a plausible incorrect value; schema/citation validity is not semantic entailment. Built-in scorer gives exact tuples, not separate per-field accuracy; derive per-field counts from saved extraction artifacts.

RQ4 — Architecture. What value is added by specialist roles, typed shared state, critic review, and memory, and what coordination cost do these mechanisms introduce?

Current evidence scope — PARTIAL: Role, free-form/typed, critic and memory ablations must retain matched tools, panels and autonomy settings. Coordination tokens, calls and latency can be measured after execution. An empty/unapproved memory store cannot establish a memory benefit.

Implementation audit's required contrast: Specialization, typed state, critic, memory and coordination costs

Implementation audit snapshot status: architectures implementable; substantive memory experiment absent. This describes implementation/design coverage at audit time, not an empirical hypothesis verdict; live execution is reported separately.

Remaining implementation or design gap: No automatic approved incident-memory writes occur in normal runs, so toggling use_memory with empty memory is a null ablation. Learned/vector memory absent.

RQ5 — Reliability. How often does the system violate hard constraints, fail to use tools correctly, or produce materially different outcomes across equivalent runs?

Current evidence scope — PARTIAL: Completed-run logs record violations, LLM/schema errors and fallback decisions. Equivalent-input repetitions, tool-validity analysis and reliability over every planned class remain separate checks; sample safety is not universal reliability.

Implementation audit's required contrast: Valid hard constraints/tools and outcome variability across equivalent runs

Implementation audit snapshot status: partially implementable. This describes implementation/design coverage at audit time, not an empirical hypothesis verdict; live execution is reported separately.

Remaining implementation or design gap: Model temperature zero/provider may ignore seeds; constant-demand variance-ratio bullwhip is undefined. Native model training seed is fixed42; simulation seeds quantify conditional variability, not independent forecast retraining. Tail regret vs an economically optimal trajectory absent.

RQ6 — Trace faithfulness. Can the decision trace be replayed and tested so that the stated evidence genuinely corresponds to the action taken?

Current evidence scope — PARTIAL: Stored sample replay and trace audit provide technical artifact evidence for their selected decisions. They do not establish causal sensitivity of every cited field, superiority over free-form rationales, or faithfulness of pending agent runs.

Implementation audit's required contrast: Replay and externally logged causal evidence validation

Implementation audit snapshot status: external audit implementable; semantic faithfulness partial. This describes implementation/design coverage at audit time, not an empirical hypothesis verdict; live execution is reported separately.

Remaining implementation or design gap: Replay reuses stored forecasts and constraints and does not re-call model or retrain. Required-artifact deletion is structural, not semantic context deletion. Narrative contradiction/rationale-to-evidence faithfulness and equivalent-prompt robustness are not supplied. B7 also produces the common structured trace. Nonbinding counterfactual factors need not change actions.

RQ7 — Human audit. Does a structured trace with lineage help reviewers detect poor decisions more quickly and with better calibrated trust than a free-form explanation?

Current evidence scope — UNRUN: No participant review-time, accuracy, trust-calibration or workload measurements are present in this automated report. Simulator approval and technical trace audit cannot answer a human-audit question.

Implementation audit's required contrast: Human audit speed/accuracy and calibrated trust

Implementation audit snapshot status: requires approved real human study. This describes implementation/design coverage at audit time, not an empirical hypothesis verdict; live execution is reported separately.

Remaining implementation or design gap: No recruitment, consent, participants or outcomes supplied. Narrative arm is deterministic summary, not an LLM rationale. Current no-lineage arm also hides state certificate, so effect confounds certificate visibility with lineage. Incorrect-constraint and poor-forecast cases need expert-vetted labels. Participant mixed-effects/power analysis absent.

Exact revised hypotheses and evidence coverage

H1. A tool-grounded multi-agent system will outperform an LLM-only policy on operating cost and reliability, while remaining close to a well-specified stochastic optimizer when the data are clean.

Current evidence scope — UNIMPLEMENTED: There is no LLM-only numerical-action policy: B6 and the multi-agent variants share deterministic tools. The first clause is unstable in the current artifact. Matched optimizer comparison can address part of the clean-state clause, without a predefined equivalence margin proving closeness.

Implementation audit's required contrast: Tool-grounded multi-agent vs LLM-only; B10 vs well-specified stochastic optimizer under clean evidence

Implementation audit snapshot status: partial; LLM-only comparator absent. This describes implementation/design coverage at audit time, not an empirical hypothesis verdict; live execution is reported separately.

Remaining implementation or design gap: B6 uses the same deterministic numerical tools and is not LLM-only. No claim about tool-grounded superiority over LLM-only is testable. B3 is native GRU/NB, not an exact DeepAR or TFT reproduction.

H2. Independent critic checks will reduce hard-constraint violations and high-cost tail events, but will increase latency and computational cost.

Current evidence scope — PARTIAL: The matched B8/B10 critic ablation can measure critic effects only after it executes; B9/B10 primarily isolates autonomy, not critic presence. Tail risk, violations, latency and compute must be reported together.

Implementation audit's required contrast: B10 minus B8: independent critic, with typed exchange and evidence gate matched

Implementation audit snapshot status: implementable with real model; unrun at required coverage. This describes implementation/design coverage at audit time, not an empirical hypothesis verdict; live execution is reported separately.

Remaining implementation or design gap: Template grounding may make semantic critic redundant; include prose and invalid-source cases. Solver holds or different extraction errors are not proof that critic prevented violations.

H3. Typed shared state will reduce grounding errors and invalid tool use relative to free-form agent communication.

Current evidence scope — PARTIAL: A matched B7/B10 ablation and grounding/tool-validity outputs are needed. Typed-schema implementation by itself is not evidence of fewer errors or a performance gain.

Implementation audit's required contrast: B8/B10 typed exchange vs B7 free-form exchange; specialist B10 vs B6 generalist

Implementation audit snapshot status: partially implementable. This describes implementation/design coverage at audit time, not an empirical hypothesis verdict; live execution is reported separately.

Remaining implementation or design gap: B7 still passes a validated deterministic final action boundary, so it does not represent wholly unconstrained free-form execution. B6 is a batched generalist and keeps shared tools. Memory needs a separate ablation.

H4. Under state-quality disturbances, evidence-gated autonomy will reduce the harmful-execution rate relative to fixed-autonomy variants. Deterministic checks will capture most simple failures; language-model reasoning will add most value where multiple sources or documents must be reconciled.

Current evidence scope — PARTIAL: Completed deterministic gates support descriptive simple-fault checks. B9/B10 isolates gating after the full matched grid. Additional language-model benefit from ambiguous cross-source reconciliation is unestablished when triage cannot apply approved state repairs.

Implementation audit's required contrast: B10 vs B9 evidence gate; B4 vs B3 deterministic gate; semantic reconciliation benefit beyond rules

Implementation audit snapshot status: gate effect implementable; semantic repair contribution not implemented. This describes implementation/design coverage at audit time, not an empirical hypothesis verdict; live execution is reported separately.

Remaining implementation or design gap: LLM triage records hypotheses but cannot repair state or clear hard failures. Current checks catch the programmed failures; no implemented ambiguous-reconciliation intervention demonstrates semantic repair benefit. Reference deviations and true violations must be reported separately.

H5. Constraint extraction will be strongest for explicit, templated rules and weaker for coupled constraints, unit conversions, and ambiguous document conventions. Deterministic validation and escalation will prevent most residual errors from reaching the optimizer.

Current evidence scope — PARTIAL: Measure template/prose and coupled/ambiguous constraint families separately using ground truth. Exact extraction, false constraints, escalation and errors reaching the optimizer need actual controlled outputs; do not inherit historical provider results.

Implementation audit's required contrast: Explicit templates vs prose, coupled MOQ, unit conversion, ambiguous conventions, and extraction errors reaching solver

Implementation audit snapshot status: controlled cases implementable; realistic corpus gap. This describes implementation/design coverage at audit time, not an empirical hypothesis verdict; live execution is reported separately.

Remaining implementation or design gap: Supplied six cases are explicit synthetic rules, not a realistic supplier/workbook or multilingual benchmark. Generic XLSX semantics and ambiguous external-document corpus are absent. Prose verification can accept a plausible incorrect value; schema/citation validity is not semantic entailment. Built-in scorer gives exact tuples, not separate per-field accuracy; derive per-field counts from saved extraction artifacts.

H6. Structured traces will achieve higher replay consistency, evidence completeness, and counterfactual sensitivity than free-form rationales.

Current evidence scope — PARTIAL: Sample replay/audit has been recorded. Structured-versus-free-form trace comparisons and planned counterfactual/deletion sensitivity must be evaluated separately before this comparative hypothesis is answered.

Implementation audit's required contrast: Structured traces vs free-form rationales on replay, evidence coverage, contradiction, counterfactual and deletion sensitivity

Implementation audit snapshot status: external evidence/tool audit implemented; rationale contrast partial. This describes implementation/design coverage at audit time, not an empirical hypothesis verdict; live execution is reported separately.

Remaining implementation or design gap: Replay reuses stored forecasts and constraints and does not re-call model or retrain. Required-artifact deletion is structural, not semantic context deletion. Narrative contradiction/rationale-to-evidence faithfulness and equivalent-prompt robustness are not supplied. B7 also produces the common structured trace. Nonbinding counterfactual factors need not change actions.

H7. Reviewers given lineage-bearing traces will identify decisions made on degraded evidence more accurately and more quickly than reviewers given only narrative explanations.

Current evidence scope — UNRUN: The reviewer experiment is unexecuted; no human accuracy/time/trust effect may be fabricated from technical traces.

Implementation audit's required contrast: Real reviewers: full lineage trace vs structured without lineage vs narrative

Implementation audit snapshot status: requires institutional approval and real participants; packet preparation only. This describes implementation/design coverage at audit time, not an empirical hypothesis verdict; live execution is reported separately.

Remaining implementation or design gap: No recruitment, consent, participants or outcomes supplied. Narrative arm is deterministic summary, not an LLM rationale. Current no-lineage arm also hides state certificate, so effect confounds certificate visibility with lineage. Incorrect-constraint and poor-forecast cases need expert-vetted labels. Participant mixed-effects/power analysis absent.

H8. Under severe demand, supply, or data shocks, bounded autonomy with escalation will produce higher safety-adjusted utility than unrestricted autonomous execution.

Current evidence scope — PARTIAL: Simulation cost, harm and escalation support conditional safety-adjusted utility. The full demand/supply/data-shock, severity and initialization design remains to be completed, and several severity controls may require implementation changes.

Implementation audit's required contrast: Bounded/gated autonomy with escalation vs fixed/unrestricted execution under severe demand, supply and data shocks

Implementation audit snapshot status: partially implementable; stress amplitude bands incomplete. This describes implementation/design coverage at audit time, not an empirical hypothesis verdict; live execution is reported separately.

Remaining implementation or design gap: B9 is fixed-full numerical-tool execution with critic, not unconstrained arbitrary ordering. Demand multiplier, lead-time shift, capacity reduction, pack change and supplier shutdown amplitudes are hardcoded. Safety weights must be fixed before evaluation and varied transparently. Hold budget changes urgency only.

Whole-experiment design coverage

The audit's proposed experiment groups are requirements and plans, not completed runs. Actual execution is counted separately from output artifacts. A completed shortened stage does not silently satisfy rolling origins, severity bands or missing baseline/ablation requirements in the revised specification.

Numerical protocol origin: The uploaded Chapter5 requires matched rolling windows and repeated replications but does not prescribe 30 seeds, 28 days or 3 origins. Candidate defaults extend existing repository settings; freeze/register chosen scope before use.

Planned group	Policies	Seeds / origins / days	Required runs	Remaining gaps
core_clean_and_stress	B1, B2, B3, B4	30 / 3 / 28	10080	See design mapping
agentic_primary	B4, B9, B10	30 / 3 / 28	7560	Injection scenario switches prose documents to template documents in run_one; isolate/label injection as template until corrected.
quality_bands_templates	B3, B4	30 / 3 / 28	7560	Most named-fault amplitude is fixed; only unit_inflation magnitude changes. Joint bands confound frequency/duration/amplitude; add one-factor-at-a-time sweeps.
quality_bands_prose	B4, B9, B10	30 / 3 / 28	11340	Exposure/rate bands only where supported; do not claim all failure amplitudes have low/medium/high implementations.
architecture_ablations	B6, B7, B8, B10	30 / 3 / 28	1440	B10 cells can be reused from matching primary settings; memory needs nonempty approved records.
foundation_clean	B5	30 / 3 / 28	90	No TFT/Moirai implementation; Chronos contamination not excluded.
forecast_scoring	—	— / — / —	—	Old origins 1800/1828 overlap main test and cannot serve as independent tuning holdout. Forecast evaluation outcomes are correlated windows, not new seed replications.
grounding	—	— / — / —	—	Realistic ambiguous/workbook corpus missing; per-field metrics require artifact postprocessing.

Planned group	Policies	Seeds / origins / days	Required runs	Remaining gaps
fixed_evidence_reliability	—	— / — / —	—	Equivalent-prompt sensitivity not supplied.
trace_audits	—	— / — / —	—	Required-artifact deletion does not test semantic causal importance; no free-form narrative counterfactual comparator.
warmup_transfer_censoring_robustness	—	— / — / —	—	Starting safety factor 1.65 is hardcoded; cannot vary through config. The named censored_demand and transshipment scenarios have no unique intervention.
human_audit	—	— / — / —	—	No recruitment, consent, participants or outcomes supplied. Narrative arm is deterministic summary, not an LLM rationale. Current no-lineage arm also hides state certificate, so effect confounds certificate visibility with lineage. Incorrect-constraint and poor-forecast cases need expert-vetted labels. Participant mixed-effects/power analysis absent.

Metrics and analysis still missing or limited

Audit snapshot: Harmful execution in code is true-constraint violation OR material distance from same-forecast clean-evidence MILP reference. It does not require an injected fault or causal attribution to corrupted evidence. Therefore legitimate clean-state classical-policy/reference differences can count as reference deviations. Fresh baseline_pilot normal rows confirm B1 has 7 deviations/0 violations and B2 has 5 deviations/0 violations. Report the two endpoints separately; do not describe all combined flags as unsafe or corruption-caused orders.

Audit snapshot: Days-of-supply decisions can be derived but are not an aggregate endpoint in daily.csv; upstream amplification is not a multi-echelon simulator.

Audit snapshot: Time-to-detect, escalation precision and autonomy calibration need explicit labels/denominators and postprocessing; precision/recall can be derived from detection.csv.

Audit snapshot: Economic tail regret vs an optimized full trajectory and forecast/constraint-labeled human counterfactuals absent.

Audit snapshot: Mixed-effects estimation and human-study power analysis absent; coupled portfolio outcomes cannot be treated as 30 independent items/stores.

Audit snapshot: All seed-paired tests require exact complete coverage; Holm multiplicity families must be explicitly defined across stages, not only independently in each generated study.

Readiness gaps recorded by the implementation audit

Audit snapshot: Local model download blocked; do not count call failures or deterministic fallbacks as LLM-backed experimental success.

Audit snapshot: B5 weights/package/revision unverified.

Audit snapshot: Existing paused10-series matched reference is supplementary; rebuild matched template controls on ALL30series.

Audit snapshot: Existing480baseline study covers only4scenarios,1origin; it is a valid subset, not wholeChapter5.

Audit snapshot: No actual human audit participants.

Compute scope — execution planning, not a throughput finding

The selected machine has an audited CPU quota of 4 cores and 32 GiB memory. Parallel workers share those resources; visible CPU count and delegated agents do not create extra inference capacity.

Planning consideration: Seed jobs can queue; one Ollama inference slot, raise timeout above measured queue+generation latency.

Planning consideration: ALL30series normal157documents, foreign_unit_moq190documents. At batch1 and28days, singleattempt≈4500–5350calls;1retry≈9000–10700. Candidate max_calls15000. Existing500 can exhaust within4days.

Planning consideration: All local-model full28day crossproducts are likely manydays or longer. Estimate using measured CPU seconds and successful call throughput; parallel agents do not increase model inference capacity. Choose comparison-driven reusable cells, retaining explicit unimplemented hypotheses.

Candidate configurations remain unregistered until selected and frozen. Their matrix sizes and call budgets are planned exposures, not successful calls, runtime estimates measured on the loaded model, or completed evidence.

Managerial adoption discussion — proposed, not an empirical finding

The revised business-school dissertation proposes an adoption sequence: establish lineage and deterministic state checks; formalize supplier and assortment constraints; introduce structured traces; use LLMs for document interpretation and anomaly triage; grant bounded autonomy only for decision classes with measured reliability. This sequence is a discussion framework, not measured organizational value, demonstrated ROI, a field deployment, or evidence that reviewers calibrated trust better.

For a manager, the relevant future comparison includes service and cost alongside harmful executions, holds, resolution delay, review workload, latency, local inference resource use and the residual error that reaches numerical tools. A conservative hold can delay replenishment and carry an operational cost. Decisions about adoption remain conditional on complete matched evidence, validation on the intended retail process and the unexecuted human-audit component.

Forecast backtests

Backtests below come from results/forecast_*.json. Training ends before each recorded test origin. Preferred holdouts at origins 1760/1788 end by index 1815, before main-study warmup at 1816. Earlier descriptive backtests at 1800/1828 overlap the main-study window and must not be used for model selection. No model or gate settings were retuned on those evaluation scores. WRMSSE, weighted scaled pinball and CRPS are scores for the selected hierarchy; full_m5_shape=false means they are not full official M5 benchmark results. Forecast accuracy alone does not establish replenishment quality or an LLM effect.

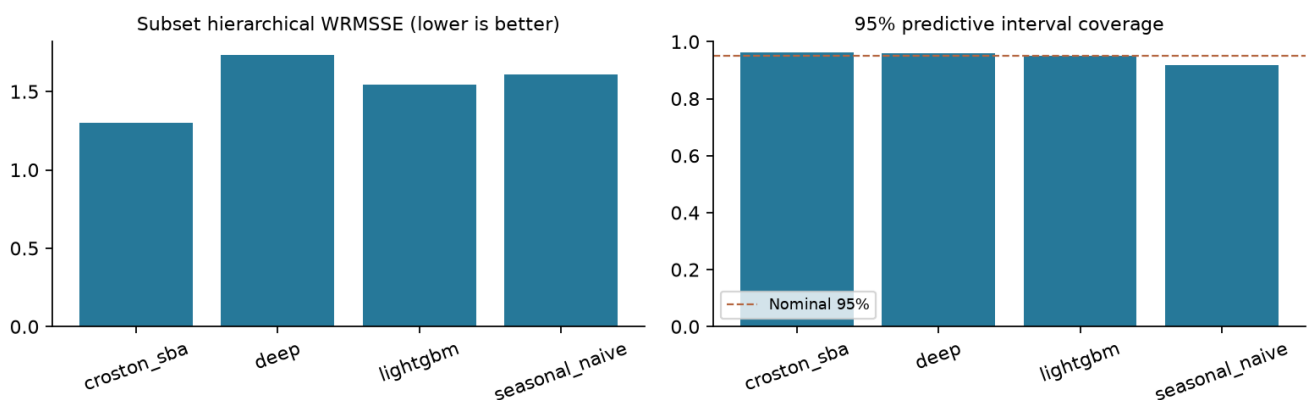
Forecast scores — pre-main-study holdout

Model	Origin / train end	Nodes	WRMSSE	Scaled pinball	CRPS	95% coverage
croston_sba	1760 / 1759	112	1.264	0.585	0.586	0.974
croston_sba	1788 / 1787	112	1.336	0.382	0.630	0.954
deep	1760 / 1759	112	1.445	0.425	0.510	0.977
deep	1788 / 1787	112	2.018	0.476	0.642	0.939
lightgbm	1760 / 1759	112	1.285	0.472	0.540	0.971
lightgbm	1788 / 1787	112	1.803	0.491	0.651	0.931
seasonal_naive	1760 / 1759	112	1.637	0.850	0.784	0.911
seasonal_naive	1788 / 1787	112	1.577	0.461	0.773	0.923

Forecast scores — overlapping descriptive backtest

Model	Origin / train end	Nodes	WRMSSE	Scaled pinball	CRPS	95% coverage
croston_sba	1800 / 1799	112	1.204	0.367	0.628	0.944
croston_sba	1828 / 1827	112	1.428	0.426	0.646	0.946
deep	1800 / 1799	112	1.796	0.407	0.582	0.939
deep	1828 / 1827	112	1.714	0.439	0.621	0.961
lightgbm	1800 / 1799	112	1.493	0.412	0.616	0.939
lightgbm	1828 / 1827	112	1.859	0.534	0.655	0.951
seasonal_naive	1800 / 1799	112	1.538	0.550	0.876	0.886
seasonal_naive	1828 / 1827	112	1.593	0.555	0.808	0.914

Actual forecast backtests; means across recorded origins, no uncertainty intervals



Means over the preferred nonoverlapping holdout origins when available. No confidence intervals or statistical significance claims are inferred from these origins.

Observed simulation outcomes

Tables include completed runs only, separated by stage and policy/scenario. n is the number of distinct independent seeds; rolling origins are averaged within seed for reported means. The repository's harmful_executions endpoint is a composite flag for true-constraint violations or material action deviation from a same-forecast clean-evidence MILP reference. The code does not require a fault or demonstrate that corrupted evidence caused the deviation. Legitimate classical policies can therefore receive reference-deviation flags in

clean normal runs. Composite flags must not all be interpreted as unsafe or corruption-caused orders; true violations and reference deviations are reported separately. Totals depend on completed exposure.

Implementation audit's harm-definition caveat: Harmful execution in code is true-constraint violation OR material distance from same-forecast clean-evidence MILP reference. It does not require an injected fault or causal attribution to corrupted evidence. Therefore legitimate clean-state classical-policy/reference differences can count as reference deviations. Fresh baseline_pilot normal rows confirm B1 has7deviations/0violations and B2 has5deviations/0violations. Report the two endpoints separately; do not describe all combined flags as unsafe or corruption-caused orders.

Partial cells can be selected by execution order and are unsuitable for policy ranking or confirmatory inference. This report makes no claim of statistical power, equivalence, significance, or treatment benefit from partial/pilot data. The repository's thirty-seed requirement is a protocol minimum, not proof of adequate power. Zero observed harms is not a universal safety guarantee.

m5_smoke — COMPLETE

Output: results/m5_smoke; config: results/run_configs/smoke_m5.yaml; dataset: data/processed/m5_smoke; series attested by completed-run manifest: 2. Seeds configured: 1; origins: 1; start index: 1800; warmup: 4; decision days: 4; approval: default with delay default; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2015-12-29 at index1795 (d_1796); decisions 2016-01-03 to 2016-01-06 at indices1800-1803 (d_1801-d_1804). Source: data/processed/m5_smoke/calendar.csv.

Policy / scenario	n seeds / runs	Mean cost	Mean fill	Harm total	True violation / ref deviation	Holds total
B1 / derived_field_collapse	1 / 1	27.137	1.000	0	0 / 0	0
B1 / feed_gap	1 / 1	19.745	1.000	0	0 / 0	0
B1 / foreign_unit_moq	1 / 1	19.745	1.000	2	2 / 0	0
B1 / normal	1 / 1	19.745	1.000	0	0 / 0	0
D0 / derived_field_collapse	1 / 1	1.806	1.000	0	0 / 0	0
D0 / feed_gap	1 / 1	2.932	1.000	0	0 / 0	0
D0 / foreign_unit_moq	1 / 1	3.495	1.000	0	0 / 0	0
D0 / normal	1 / 1	3.495	1.000	0	0 / 0	0
D1 / derived_field_collapse	1 / 1	0.961	1.000	0	0 / 0	3
D1 / feed_gap	1 / 1	0.961	1.000	0	0 / 0	3
D1 / foreign_unit_moq	1 / 1	3.495	1.000	0	0 / 0	0
D1 / normal	1 / 1	3.495	1.000	0	0 / 0	0

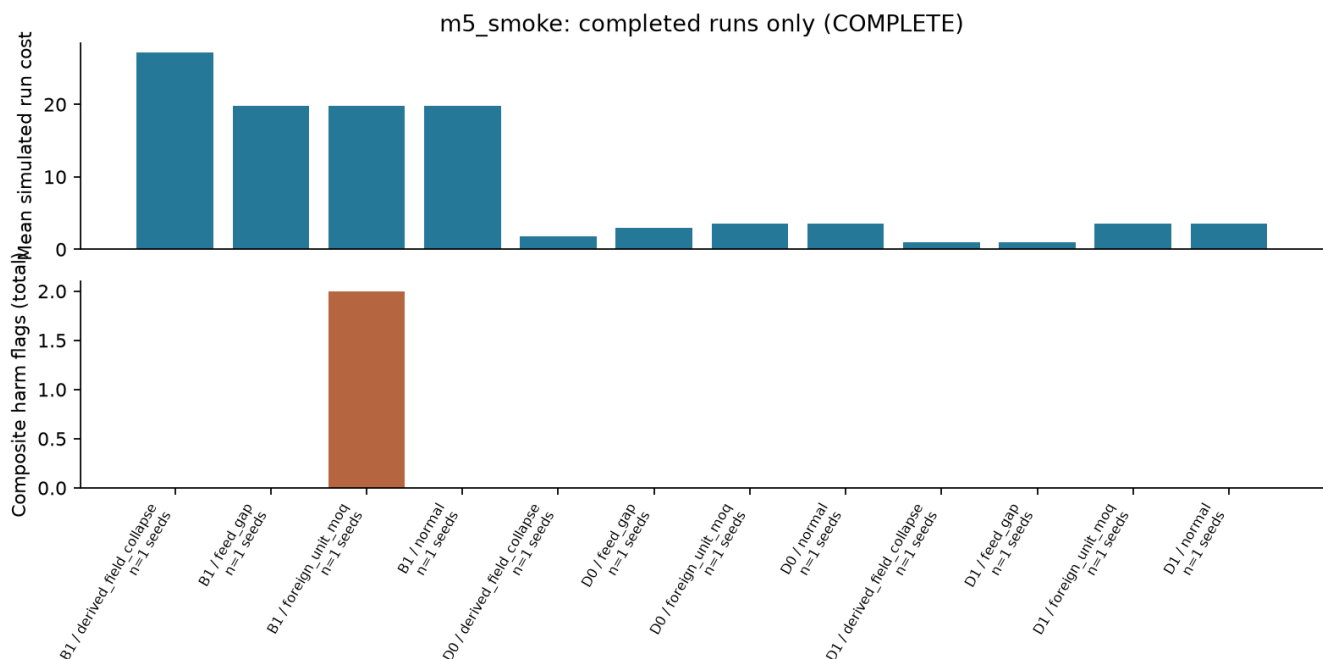
Policy / scenario	LLM calls	LLM errors	Fallback decisions
B1 / derived_field_collapse	0	0	0
B1 / feed_gap	0	0	0
B1 / foreign_unit_moq	0	0	0
B1 / normal	0	0	0
D0 / derived_field_collapse	0	0	0
D0 / feed_gap	0	0	0
D0 / foreign_unit_moq	0	0	0
D0 / normal	0	0	0

Policy / scenario	LLM calls	LLM errors	Fallback decisions
D1 / derived_field_collapse	0	0	0
D1 / feed_gap	0	0	0
D1 / foreign_unit_moq	0	0	0
D1 / normal	0	0	0

Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
B1 / derived_field_collapse	1.000	0.000	4.125	13.000	0.615
B1 / feed_gap	1.000	0.000	1.500	13.000	0.615
B1 / foreign_unit_moq	1.000	0.000	1.500	13.000	0.615
B1 / normal	1.000	0.000	1.500	13.000	0.615
D0 / derived_field_collapse	1.000	0.000	0.000	12.250	0.653
D0 / feed_gap	1.000	0.000	0.000	11.250	0.711
D0 / foreign_unit_moq	1.000	0.000	0.000	10.750	0.744
D0 / normal	1.000	0.000	0.000	10.750	0.744
D1 / derived_field_collapse	1.000	0.000	0.000	13.000	0.615
D1 / feed_gap	1.000	0.000	0.000	13.000	0.615
D1 / foreign_unit_moq	1.000	0.000	0.000	10.750	0.744
D1 / normal	1.000	0.000	0.000	10.750	0.744

Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

The repository also wrote study_summary.json with replication-level cost/CVaR and utility. It is preserved in report_data.json. It may lag live worker results; this report's tables are refreshed from complete per-run artifacts. Run-level daily CVaR is not substituted for replication-level tail risk.



Descriptive simulated outcomes from completed cells; exposure and panel differ across stages. The harmful execution count combines distinct mechanisms detailed in the table.

ml_m5_smoke — COMPLETE

Output: results/ml_m5_smoke; config: results/run_configs/ml_m5_smoke.yaml; dataset: data/processed/m5_smoke; series attested by completed-run manifest: 2. Seeds configured: 1; origins: 1; start index: 1800; warmup: 4; decision days: 4; approval: default with delay default; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2015-12-29 at index1795 (d_1796); decisions 2016-01-03 to 2016-01-06 at indices1800-1803 (d_1801-d_1804). Source: data/processed/m5_smoke/calendar.csv.

Policy / scenario	n seeds / runs	Mean cost	Mean fill	Harm total	True violation / ref deviation	Holds total
B1 / derived_field_collapse	1 / 1	27.137	1.000	0	0 / 0	0
B1 / feed_gap	1 / 1	19.745	1.000	0	0 / 0	0
B1 / foreign_unit_moq	1 / 1	19.745	1.000	2	2 / 0	0
B1 / normal	1 / 1	19.745	1.000	0	0 / 0	0
B2 / derived_field_collapse	1 / 1	10.353	1.000	0	0 / 0	0
B2 / feed_gap	1 / 1	0.961	1.000	0	0 / 0	0
B2 / foreign_unit_moq	1 / 1	0.961	1.000	0	0 / 0	0
B2 / normal	1 / 1	0.961	1.000	0	0 / 0	0
B3 / derived_field_collapse	1 / 1	12.042	1.000	0	0 / 0	0
B3 / feed_gap	1 / 1	3.776	1.000	0	0 / 0	0
B3 / foreign_unit_moq	1 / 1	3.776	1.000	0	0 / 0	0
B3 / normal	1 / 1	3.776	1.000	0	0 / 0	0
B4 / derived_field_collapse	1 / 1	2.650	1.000	0	0 / 0	3
B4 / feed_gap	1 / 1	2.650	1.000	0	0 / 0	3
B4 / foreign_unit_moq	1 / 1	3.776	1.000	0	0 / 0	0
B4 / normal	1 / 1	3.776	1.000	0	0 / 0	0

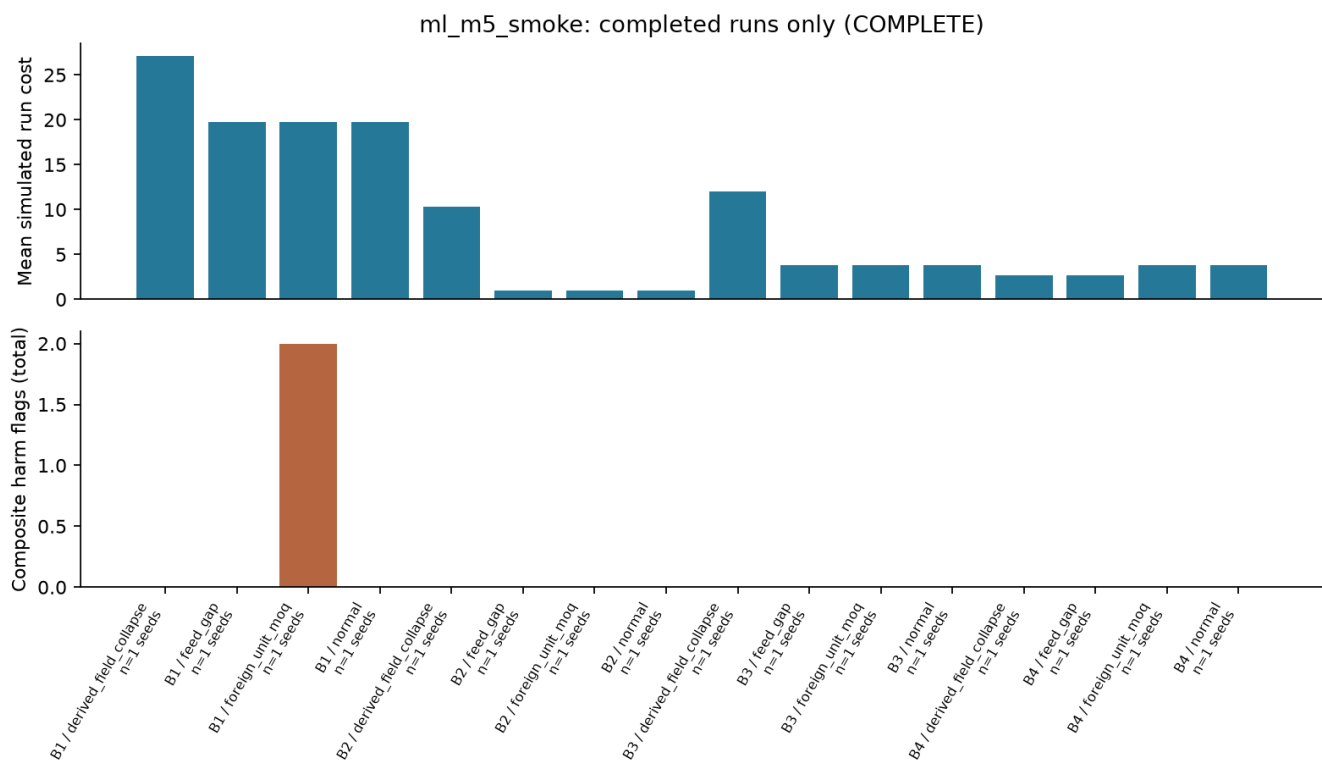
Policy / scenario	LLM calls	LLM errors	Fallback decisions
B1 / derived_field_collapse	0	0	0
B1 / feed_gap	0	0	0
B1 / foreign_unit_moq	0	0	0
B1 / normal	0	0	0
B2 / derived_field_collapse	0	0	0
B2 / feed_gap	0	0	0
B2 / foreign_unit_moq	0	0	0
B2 / normal	0	0	0
B3 / derived_field_collapse	0	0	0
B3 / feed_gap	0	0	0
B3 / foreign_unit_moq	0	0	0
B3 / normal	0	0	0
B4 / derived_field_collapse	0	0	0

Policy / scenario	LLM calls	LLM errors	Fallback decisions
B4 / feed_gap	0	0	0
B4 / foreign_unit_moq	0	0	0
B4 / normal	0	0	0

Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
B1 / derived_field_collapse	1.000	0.000	4.125	13.000	0.615
B1 / feed_gap	1.000	0.000	1.500	13.000	0.615
B1 / foreign_unit_moq	1.000	0.000	1.500	13.000	0.615
B1 / normal	1.000	0.000	1.500	13.000	0.615
B2 / derived_field_collapse	1.000	0.000	1.125	13.000	0.615
B2 / feed_gap	1.000	0.000	0.000	13.000	0.615
B2 / foreign_unit_moq	1.000	0.000	0.000	13.000	0.615
B2 / normal	1.000	0.000	0.000	13.000	0.615
B3 / derived_field_collapse	1.000	0.000	1.125	11.500	0.696
B3 / feed_gap	1.000	0.000	0.000	10.500	0.762
B3 / foreign_unit_moq	1.000	0.000	0.000	10.500	0.762
B3 / normal	1.000	0.000	0.000	10.500	0.762
B4 / derived_field_collapse	1.000	0.000	0.000	11.500	0.696
B4 / feed_gap	1.000	0.000	0.000	11.500	0.696
B4 / foreign_unit_moq	1.000	0.000	0.000	10.500	0.762
B4 / normal	1.000	0.000	0.000	10.500	0.762

Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

The repository also wrote study_summary.json with replication-level cost/CVaR and utility. It is preserved in report_data.json. It may lag live worker results; this report's tables are refreshed from complete per-run artifacts. Run-level daily CVaR is not substituted for replication-level tail risk.



Descriptive simulated outcomes from completed cells; exposure and panel differ across stages. The harmful execution count combines distinct mechanisms detailed in the table.

baseline_pilot — COMPLETE

Output: results/baseline_pilot; config: results/run_configs/baseline_pilot.yaml; dataset: data/processed/m5; series attested by completed-run manifest: 30. Seeds configured: 2; origins: 1; start index: 1830; warmup: 14; decision days: 7; approval: oracle with delay 1; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2016-01-18 at index1815 (d_1816); decisions 2016-02-02 to 2016-02-08 at indices1830-1836 (d_1831-d_1837). Source: data/processed/m5/calendar.csv.

Policy / scenario	n seeds / runs	Mean cost	Mean fill	Harm total	True violation / ref deviation	Holds total
B1 / derived_field_collapse	2 / 2	731.757	0.934	10	0 / 10	0
B1 / normal	2 / 2	390.874	0.932	14	0 / 14	0
B2 / derived_field_collapse	2 / 2	625.233	0.911	6	0 / 6	0
B2 / normal	2 / 2	388.219	0.802	10	0 / 10	0
B3 / derived_field_collapse	2 / 2	1,051.913	0.980	7	0 / 7	0
B3 / normal	2 / 2	562.691	0.980	0	0 / 0	0
B4 / derived_field_collapse	2 / 2	525.882	0.941	1	0 / 1	8
B4 / normal	2 / 2	554.447	0.952	0	0 / 0	2

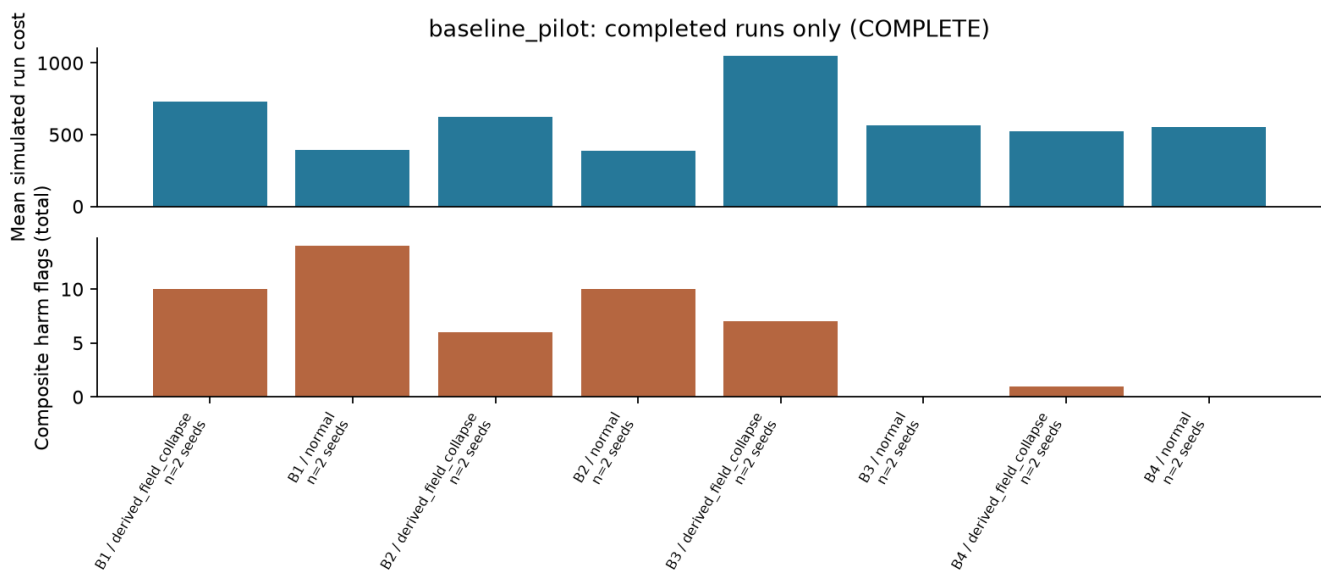
Policy / scenario	LLM calls	LLM errors	Fallback decisions
B1 / derived_field_collapse	0	0	0
B1 / normal	0	0	0
B2 / derived_field_collapse	0	0	0

Policy / scenario	LLM calls	LLM errors	Fallback decisions
B2 / normal	0	0	0
B3 / derived_field_collapse	0	0	0
B3 / normal	0	0	0
B4 / derived_field_collapse	0	0	0
B4 / normal	0	0	0

Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
B1 / derived_field_collapse	0.643	0.014	86.022	323.429	0.635
B1 / normal	0.571	0.017	1.570	237.000	0.867
B2 / derived_field_collapse	0.429	0.021	77.251	282.643	0.711
B2 / normal	0.214	0.057	2.965	196.857	0.901
B3 / derived_field_collapse	0.786	0.010	124.693	429.786	0.502
B3 / normal	0.786	0.010	25.819	291.286	0.745
B4 / derived_field_collapse	0.571	0.017	58.577	244.214	0.849
B4 / normal	0.643	0.014	31.919	271.214	0.773

Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

The repository also wrote study_summary.json with replication-level cost/CVaR and utility. It is preserved in report_data.json. It may lag live worker results; this report's tables are refreshed from complete per-run artifacts. Run-level daily CVaR is not substituted for replication-level tail risk.



Descriptive simulated outcomes from completed cells; exposure and panel differ across stages. The harmful execution count combines distinct mechanisms detailed in the table.

main_study_stage1 — PARTIAL

Output: results/main_study_stage1; config: configs/main_study_stage1.yaml; dataset: data/processed/m5; series attested by completed-run manifest: 30. Seeds configured: 30; origins: 1; start index: 1830; warmup: 14; decision days: 28; approval: oracle with delay 1; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2016-01-18 at index1815 (d_1816); decisions 2016-02-02 to 2016-02-29 at indices1830-1857 (d_1831-d_1858). Source: data/processed/m5/calendar.csv.

Policy / scenario	n seeds / runs	Mean cost	Mean fill	Harm total	True violation / ref deviation	Holds total
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B1 / normal	20 / 20	1,457.503	0.949	559	0 / 559	0
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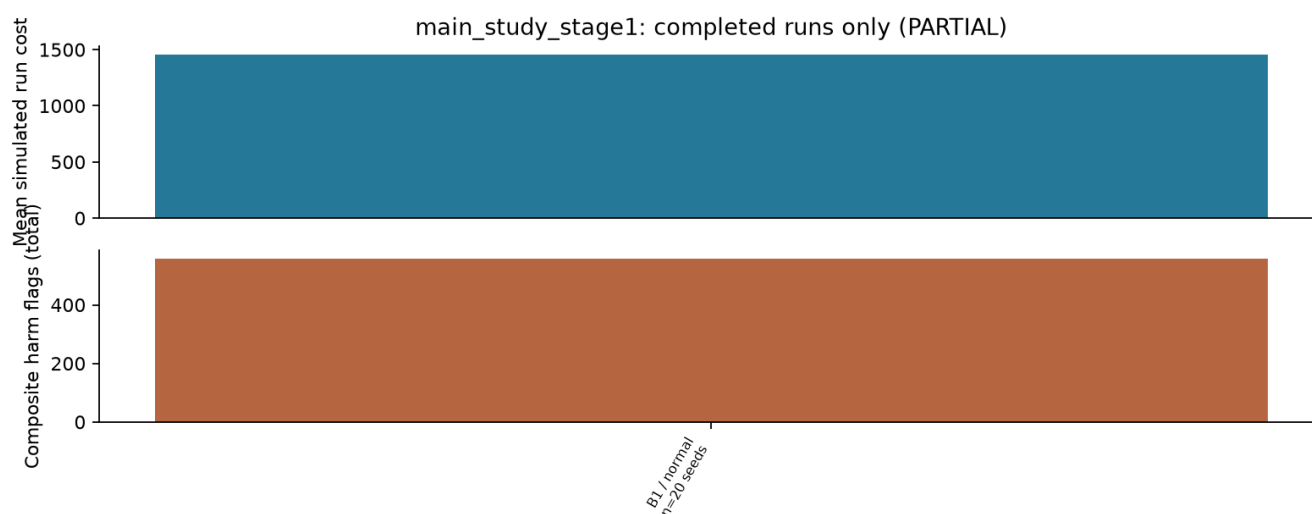
Policy / scenario	LLM calls	LLM errors	Fallback decisions
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B1 / normal	0	0	0
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Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
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B1 / normal	0.623	0.017	1.647	251.787	3.153
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Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.



Descriptive simulated outcomes from completed cells; exposure and panel differ across stages. The harmful execution count combines distinct mechanisms detailed in the table.

foundation_30series — BLOCKED

Output: results/foundation_30series; config: results/run_configs/foundation_30series.yaml; dataset: data/processed/m5; series attested by completed-run manifest: no completed-run manifest yet. Seeds configured: 30; origins: 3; start index: 1830; warmup: 14; decision days: 28; approval: oracle with delay 1; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2016-01-18 at index1815 (d_1816); decisions 2016-02-02 to 2016-02-29 at indices1830-1857 (d_1831-d_1858). Source: data/processed/m5/calendar.csv.

Historical origin 1: training data end 2016-02-15 at index1843 (d_1844); decisions 2016-03-01 to 2016-03-28 at indices1858-1885 (d_1859-d_1886). Source: data/processed/m5/calendar.csv.

Historical origin 2: training data end 2016-03-14 at index1871 (d_1872); decisions 2016-03-29 to 2016-04-25 at indices1886-1913 (d_1887-d_1914). Source: data/processed/m5/calendar.csv.

No completed observations are available for this table.

No completed observations are available for this table.

No completed observations are available for this table.

Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

llm_architectures_ollama — BLOCKED

Output: results/llm_architectures_ollama; config: results/run_configs/llm_architectures_ollama.yaml; dataset: data/processed/m5_smoke; series attested by completed-run manifest: no completed-run manifest yet. Seeds configured: 1; origins: 1; start index: 1800; warmup: 7; decision days: 4; approval: oracle with delay 1; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2015-12-26 at index1792 (d_1793); decisions 2016-01-03 to 2016-01-06 at indices1800-1803 (d_1801-d_1804). Source: data/processed/m5_smoke/calendar.csv.

No completed observations are available for this table.

No completed observations are available for this table.

No completed observations are available for this table.

Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

llm_prose_reference_30series — PARTIAL

Output: results/llm_prose_reference_30series; config: results/run_configs/llm_prose_reference_30series.yaml; dataset: data/processed/m5; series attested by completed-run manifest: 30. Seeds configured: 30; origins: 3; start index: 1830; warmup: 14; decision days: 28; approval: oracle with delay 1; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2016-01-18 at index1815 (d_1816); decisions 2016-02-02 to 2016-02-29 at indices1830-1857 (d_1831-d_1858). Source: data/processed/m5/calendar.csv.

Historical origin 1: training data end 2016-02-15 at index1843 (d_1844); decisions 2016-03-01 to 2016-03-28 at indices1858-1885 (d_1859-d_1886). Source: data/processed/m5/calendar.csv.

Historical origin 2: training data end 2016-03-14 at index1871 (d_1872); decisions 2016-03-29 to 2016-04-25 at indices1886-1913 (d_1887-d_1914). Source: data/processed/m5/calendar.csv.

Policy / scenario	n seeds / runs	Mean cost	Mean fill	Harm total	True violation / ref deviation	Holds total
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B3 / normal	1 / 1	1,630.796	0.982	0	0 / 0	0
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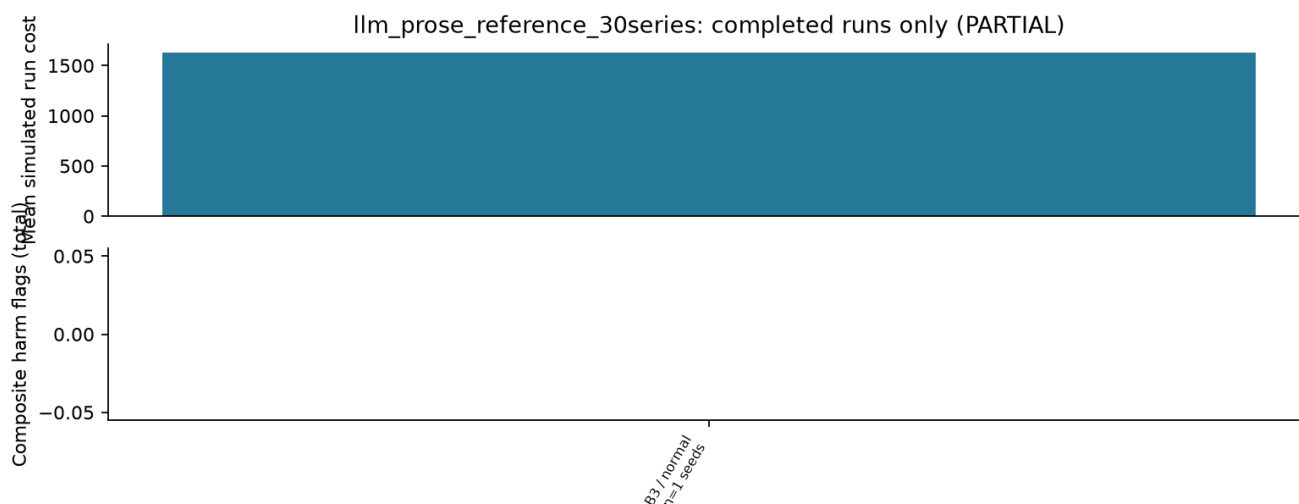
Policy / scenario	LLM calls	LLM errors	Fallback decisions
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B3 / normal	0	0	0
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Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
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B3 / normal	0.821	0.007	2.785	293.393	2.778
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Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.



Descriptive simulated outcomes from completed cells; exposure and panel differ across stages. The harmful execution count combines distinct mechanisms detailed in the table.

llm_prose_study_ollama — BLOCKED

Output: results/llm_prose_study_ollama; config: results/run_configs/llm_prose_study_ollama.yaml; dataset: data/processed/m5; series attested by completed-run manifest: no completed-run manifest yet. Seeds configured: 30; origins: 3; start index: 1830; warmup: 14; decision days: 28; approval: oracle with delay 1; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2016-01-18 at index1815 (d_1816); decisions 2016-02-02 to 2016-02-29 at indices1830-1857 (d_1831-d_1858). Source: data/processed/m5/calendar.csv.

Historical origin 1: training data end 2016-02-15 at index1843 (d_1844); decisions 2016-03-01 to 2016-03-28 at indices1858-1885 (d_1859-d_1886). Source: data/processed/m5/calendar.csv.

Historical origin 2: training data end 2016-03-14 at index1871 (d_1872); decisions 2016-03-29 to 2016-04-25 at indices1886-1913 (d_1887-d_1914). Source: data/processed/m5/calendar.csv.

No completed observations are available for this table.

No completed observations are available for this table.

No completed observations are available for this table.

Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

llm_prose_study_reference — COMPLETE

Output: results/llm_prose_study_reference; config: configs/llm_prose_study_reference.yaml; dataset: data/processed/m5_item1; series attested by completed-run manifest: 10. Seeds configured: 30; origins: 1; start index: 1830; warmup: 14; decision days: 4; approval: oracle with delay 1; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2016-01-18 at index1815 (d_1816); decisions 2016-02-02 to 2016-02-05 at indices1830-1833 (d_1831-d_1834). Source: data/processed/m5_item1/calendar.csv.

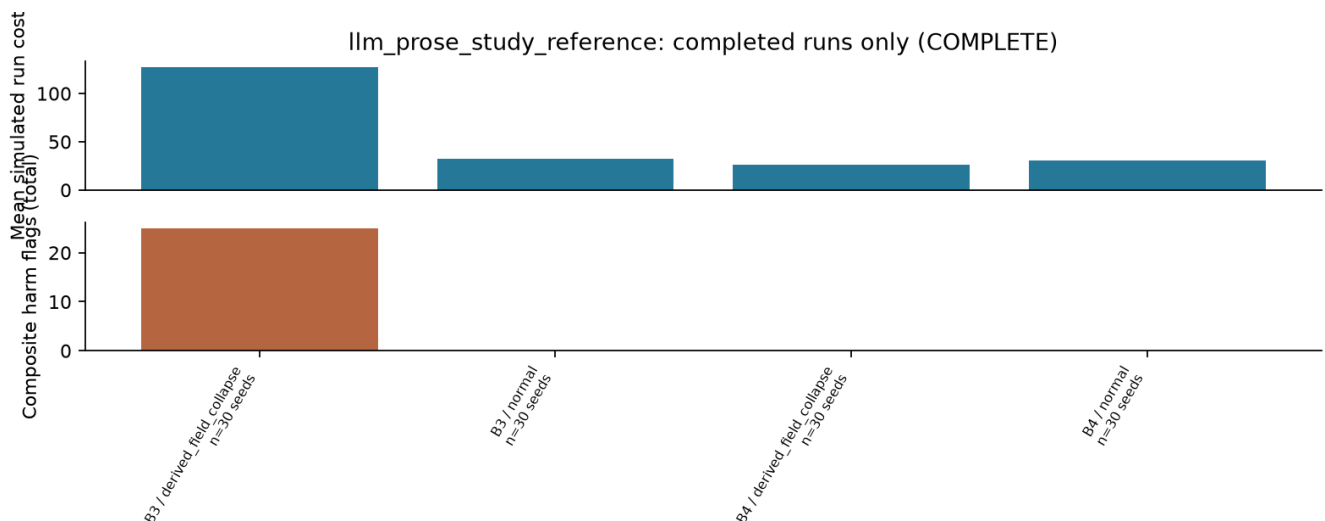
Policy / scenario	n seeds / runs	Mean cost	Mean fill	Harm total	True violation / ref deviation	Holds total
B3 / derived_field_collapse	30 / 30	126.973	1.000	25	0 / 25	0
B3 / normal	30 / 30	32.024	1.000	0	0 / 0	0
B4 / derived_field_collapse	30 / 30	25.774	1.000	0	0 / 0	60
B4 / normal	30 / 30	30.834	1.000	0	0 / 0	2

Policy / scenario	LLM calls	LLM errors	Fallback decisions
B3 / derived_field_collapse	0	0	0
B3 / normal	0	0	0
B4 / derived_field_collapse	0	0	0
B4 / normal	0	0	0

Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
B3 / derived_field_collapse	1.000	0.000	250.680	99.008	0.144
B3 / normal	1.000	0.000	2.951	85.408	0.167
B4 / derived_field_collapse	1.000	0.000	3.044	87.992	0.162
B4 / normal	1.000	0.000	2.396	85.508	0.166

Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

The repository also wrote study_summary.json with replication-level cost/CVaR and utility. It is preserved in report_data.json. It may lag live worker results; this report's tables are refreshed from complete per-run artifacts. Run-level daily CVaR is not substituted for replication-level tail risk.



Descriptive simulated outcomes from completed cells; exposure and panel differ across stages. The harmful execution count combines distinct mechanisms detailed in the table.

llm_template_architectures_30series — BLOCKED

Output: results/llm_template_architectures_30series; config: results/run_configs/llm_template_architectures_30series.yaml; dataset: data/processed/m5; series attested by completed-run manifest: no completed-run manifest yet. Seeds configured: 30; origins: 3; start index: 1830;

warmup: 14; decision days: 28; approval: oracle with delay 1; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2016-01-18 at index1815 (d_1816); decisions 2016-02-02 to 2016-02-29 at indices1830-1857 (d_1831-d_1858). Source: data/processed/m5/calendar.csv.

Historical origin 1: training data end 2016-02-15 at index1843 (d_1844); decisions 2016-03-01 to 2016-03-28 at indices1858-1885 (d_1859-d_1886). Source: data/processed/m5/calendar.csv.

Historical origin 2: training data end 2016-03-14 at index1871 (d_1872); decisions 2016-03-29 to 2016-04-25 at indices1886-1913 (d_1887-d_1914). Source: data/processed/m5/calendar.csv.

No completed observations are available for this table.

No completed observations are available for this table.

No completed observations are available for this table.

Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

parallel_runner_validation — COMPLETE

Output: results/parallel_runner_validation; config: results/run_configs/parallel_runner_validation.yaml; dataset: data/processed/m5_smoke; series attested by completed-run manifest: 2. Seeds configured: 2; origins: 1; start index: 1800; warmup: 4; decision days: 1; approval: default with delay default; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2015-12-29 at index1795 (d_1796); decisions 2016-01-03 to 2016-01-03 at indices1800-1800 (d_1801-d_1801). Source: data/processed/m5_smoke/calendar.csv.

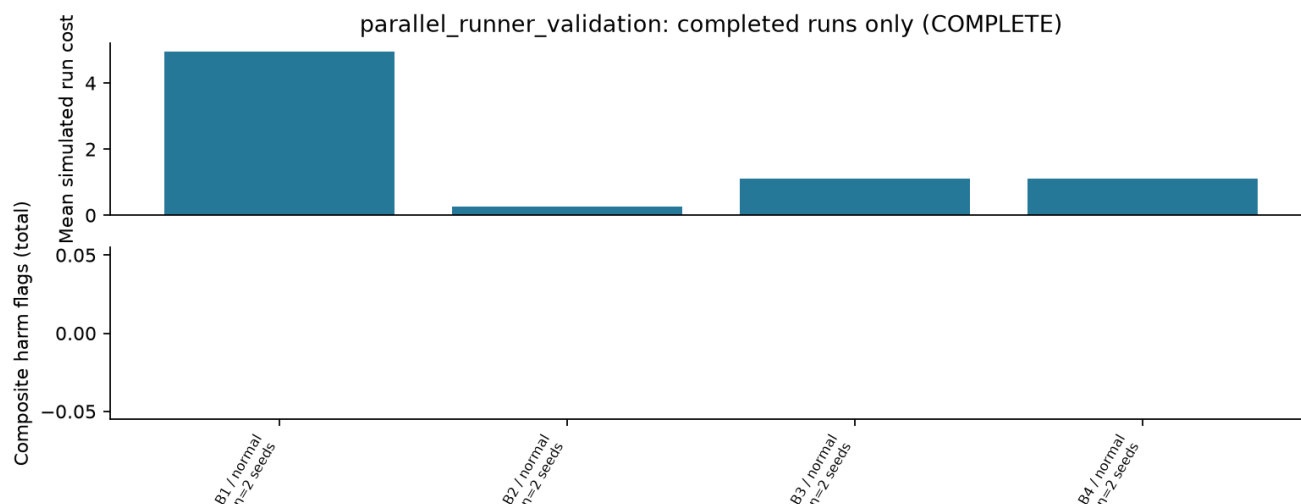
Policy / scenario	n seeds / runs	Mean cost	Mean fill	Harm total	True violation / ref deviation	Holds total
B1 / normal	2 / 2	4.955	1.000	0	0 / 0	0
B2 / normal	2 / 2	0.259	1.000	0	0 / 0	0
B3 / normal	2 / 2	1.103	1.000	0	0 / 0	0
B4 / normal	2 / 2	1.103	1.000	0	0 / 0	0

Policy / scenario	LLM calls	LLM errors	Fallback decisions
B1 / normal	0	0	0
B2 / normal	0	0	0
B3 / normal	0	0	0
B4 / normal	0	0	0

Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
B1 / normal	1.000	0.000	—	14.000	0.000
B2 / normal	1.000	0.000	—	14.000	0.000
B3 / normal	1.000	0.000	—	11.000	0.000
B4 / normal	1.000	0.000	—	11.000	0.000

Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

The repository also wrote study_summary.json with replication-level cost/CVaR and utility. It is preserved in report_data.json. It may lag live worker results; this report's tables are refreshed from complete per-run artifacts. Run-level daily CVaR is not substituted for replication-level tail risk.



Descriptive simulated outcomes from completed cells; exposure and panel differ across stages. The harmful execution count combines distinct mechanisms detailed in the table.

six_hour_agentic_prose — COMPLETE

Output: results/six_hour_agentic_prose; config: results/run_configs/six_hour_agentic_prose.yaml; dataset: data/processed/m5; series attested by completed-run manifest: 30. Seeds configured: 1; origins: 1; start index: 1830; warmup: 14; decision days: 2; approval: oracle with delay 1; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2016-01-18 at index1815 (d_1816); decisions 2016-02-02 to 2016-02-03 at indices1830-1831 (d_1831-d_1832). Source: data/processed/m5/calendar.csv.

B3/B4 in this prose stage are intentional deterministic-parser comparators. The parser requires the structured RULE carrier and recognizes no rules in controlled prose, so missing required constraint coverage causes holds. They are not structured-rule oracle benchmarks. Separate registered two-day template controls are shown under their own stage; the seven-day numerical template costs are not pooled into this comparison.

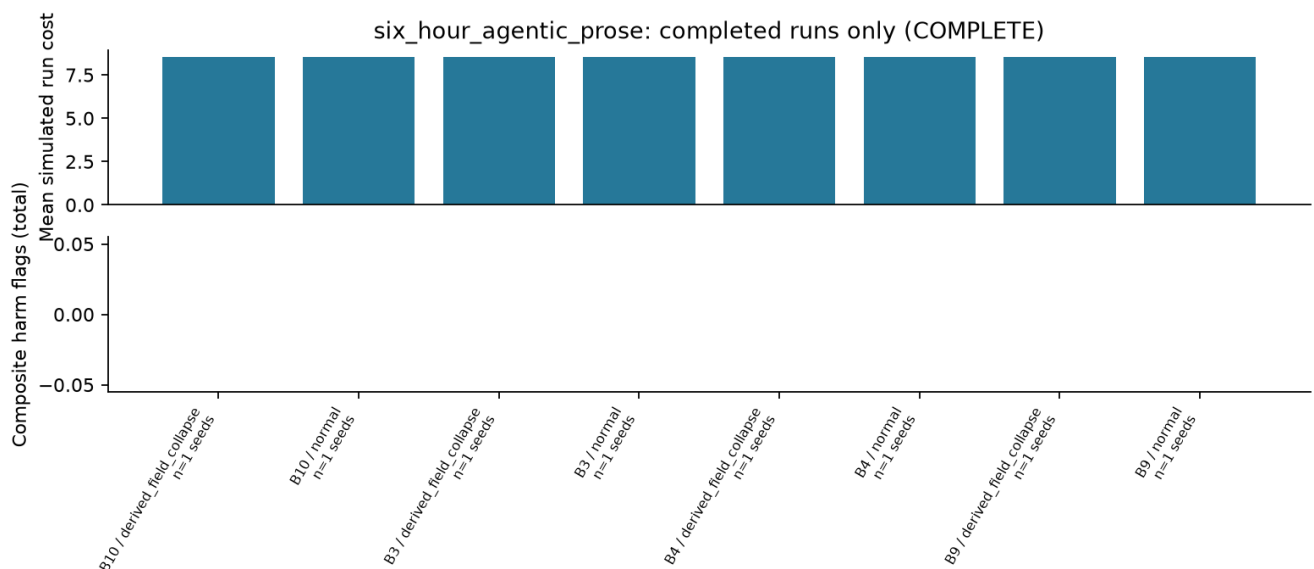
Policy / scenario	n seeds / runs	Mean cost	Mean fill	Harm total	True violation / ref deviation	Holds total
B10 / derived_field_collapse	1 / 1	8.513	1.000	0	0 / 0	2
B10 / normal	1 / 1	8.513	1.000	0	0 / 0	2
B3 / derived_field_collapse	1 / 1	8.513	1.000	0	0 / 0	2
B3 / normal	1 / 1	8.513	1.000	0	0 / 0	2
B4 / derived_field_collapse	1 / 1	8.513	1.000	0	0 / 0	2
B4 / normal	1 / 1	8.513	1.000	0	0 / 0	2
B9 / derived_field_collapse	1 / 1	8.513	1.000	0	0 / 0	2
B9 / normal	1 / 1	8.513	1.000	0	0 / 0	2

Policy / scenario	LLM calls	LLM errors	Fallback decisions
B10 / derived_field_collapse	5	0	0
B10 / normal	8	0	0
B3 / derived_field_collapse	0	0	0
B3 / normal	0	0	0
B4 / derived_field_collapse	0	0	0
B4 / normal	0	0	0
B9 / derived_field_collapse	9	0	0
B9 / normal	8	0	0

Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
B10 / derived_field_collapse	1.000	0.000	0.000	237.500	0.232
B10 / normal	1.000	0.000	0.000	237.500	0.232
B3 / derived_field_collapse	1.000	0.000	0.000	237.500	0.232
B3 / normal	1.000	0.000	0.000	237.500	0.232
B4 / derived_field_collapse	1.000	0.000	0.000	237.500	0.232
B4 / normal	1.000	0.000	0.000	237.500	0.232
B9 / derived_field_collapse	1.000	0.000	0.000	237.500	0.232
B9 / normal	1.000	0.000	0.000	237.500	0.232

Completed-run recorded tokens=112,739; LLM calls=30; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

The repository also wrote study_summary.json with replication-level cost/CVaR and utility. It is preserved in report_data.json. It may lag live worker results; this report's tables are refreshed from complete per-run artifacts. Run-level daily CVaR is not substituted for replication-level tail risk.



Descriptive simulated outcomes from completed cells; exposure and panel differ across stages. The harmful execution count combines distinct mechanisms detailed in the table.

six_hour_numeric — COMPLETE

Output: results/six_hour_numeric; config: results/run_configs/six_hour_numeric.yaml; dataset: data/processed/m5; series attested by completed-run manifest: 30. Seeds configured: 8; origins: 1; start index: 1830; warmup: 14; decision days: 7; approval: oracle with delay 1; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2016-01-18 at index1815 (d_1816); decisions 2016-02-02 to 2016-02-08 at indices1830-1836 (d_1831-d_1837). Source: data/processed/m5/calendar.csv.

Policy / scenario	n seeds / runs	Mean cost	Mean fill	Harm total	True violation / ref deviation	Holds total
B1 / derived_field_collapse	8 / 8	624.865	0.935	46	0 / 46	0
B1 / feed_gap	8 / 8	450.162	0.908	56	0 / 56	0
B1 / normal	8 / 8	394.844	0.931	55	0 / 55	0
B2 / derived_field_collapse	8 / 8	513.635	0.876	34	0 / 34	0
B2 / feed_gap	8 / 8	416.981	0.788	34	0 / 34	0
B2 / normal	8 / 8	380.920	0.827	44	0 / 44	0
B3 / derived_field_collapse	8 / 8	493.214	0.930	22	0 / 22	0
B3 / feed_gap	8 / 8	415.272	0.905	27	19 / 8	0
B3 / normal	8 / 8	363.719	0.946	1	0 / 1	0
B4 / derived_field_collapse	8 / 8	380.184	0.890	1	0 / 1	24
B4 / feed_gap	8 / 8	376.781	0.892	1	0 / 1	24
B4 / normal	8 / 8	360.245	0.947	3	0 / 3	0

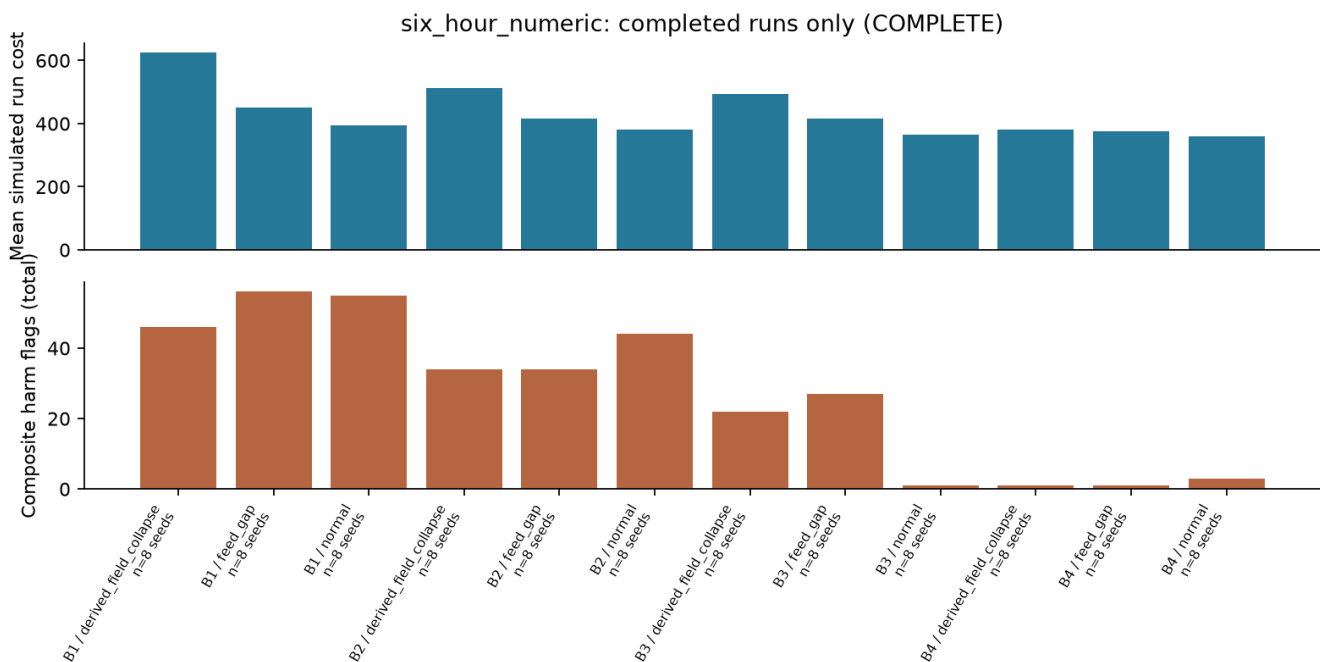
Policy / scenario	LLM calls	LLM errors	Fallback decisions
B1 / derived_field_collapse	0	0	0
B1 / feed_gap	0	0	0
B1 / normal	0	0	0
B2 / derived_field_collapse	0	0	0
B2 / feed_gap	0	0	0
B2 / normal	0	0	0
B3 / derived_field_collapse	0	0	0
B3 / feed_gap	0	0	0
B3 / normal	0	0	0
B4 / derived_field_collapse	0	0	0
B4 / feed_gap	0	0	0
B4 / normal	0	0	0

Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
B1 / derived_field_collapse	0.554	0.022	60.293	261.179	0.794
B1 / feed_gap	0.339	0.032	2.914	209.982	0.955
B1 / normal	0.464	0.025	2.138	210.804	0.979
B2 / derived_field_collapse	0.411	0.040	51.512	227.375	0.852

Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
B2 / feed_gap	0.196	0.067	5.480	181.571	0.966
B2 / normal	0.250	0.059	2.361	184.500	0.994
B3 / derived_field_collapse	0.500	0.025	19.551	217.750	0.953
B3 / feed_gap	0.357	0.040	2.256	175.714	1.136
B3 / normal	0.554	0.021	4.111	171.536	1.225
B4 / derived_field_collapse	0.393	0.041	9.260	173.143	1.137
B4 / feed_gap	0.411	0.040	9.181	173.214	1.139
B4 / normal	0.518	0.021	4.249	170.768	1.231

Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

The repository also wrote study_summary.json with replication-level cost/CVaR and utility. It is preserved in report_data.json. It may lag live worker results; this report's tables are refreshed from complete per-run artifacts. Run-level daily CVaR is not substituted for replication-level tail risk.



Descriptive simulated outcomes from completed cells; exposure and panel differ across stages. The harmful execution count combines distinct mechanisms detailed in the table.

six_hour_numeric_extension — COMPLETE

Output: results/six_hour_numeric_extension; config: results/run_configs/six_hour_numeric_extension.yaml; dataset: data/processed/m5; series attested by completed-run manifest: 30. Seeds configured: 22; origins: 1; start index: 1830; warmup: 14; decision days: 7; approval: oracle with delay 1; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2016-01-18 at index1815 (d_1816); decisions 2016-02-02 to 2016-02-08 at indices1830-1836 (d_1831-d_1837). Source: data/processed/m5/calendar.csv.

Policy / scenario	n seeds / runs	Mean cost	Mean fill	Harm total	True violation / ref deviation	Holds total
B1 / derived_field_collapse	22 / 22	619.686	0.925	133	0 / 133	0
B1 / feed_gap	22 / 22	440.646	0.909	154	0 / 154	0
B1 / normal	22 / 22	401.666	0.917	154	0 / 154	0
B2 / derived_field_collapse	22 / 22	561.821	0.843	88	0 / 88	0
B2 / feed_gap	22 / 22	448.461	0.767	97	0 / 97	0
B2 / normal	22 / 22	406.433	0.804	117	0 / 117	0
B3 / derived_field_collapse	22 / 22	498.439	0.940	56	0 / 56	0
B3 / feed_gap	22 / 22	429.314	0.902	73	44 / 29	0
B3 / normal	22 / 22	379.595	0.937	9	0 / 9	0
B4 / derived_field_collapse	22 / 22	365.509	0.911	5	0 / 5	66
B4 / feed_gap	22 / 22	364.529	0.910	4	0 / 4	66
B4 / normal	22 / 22	379.002	0.937	8	0 / 8	0

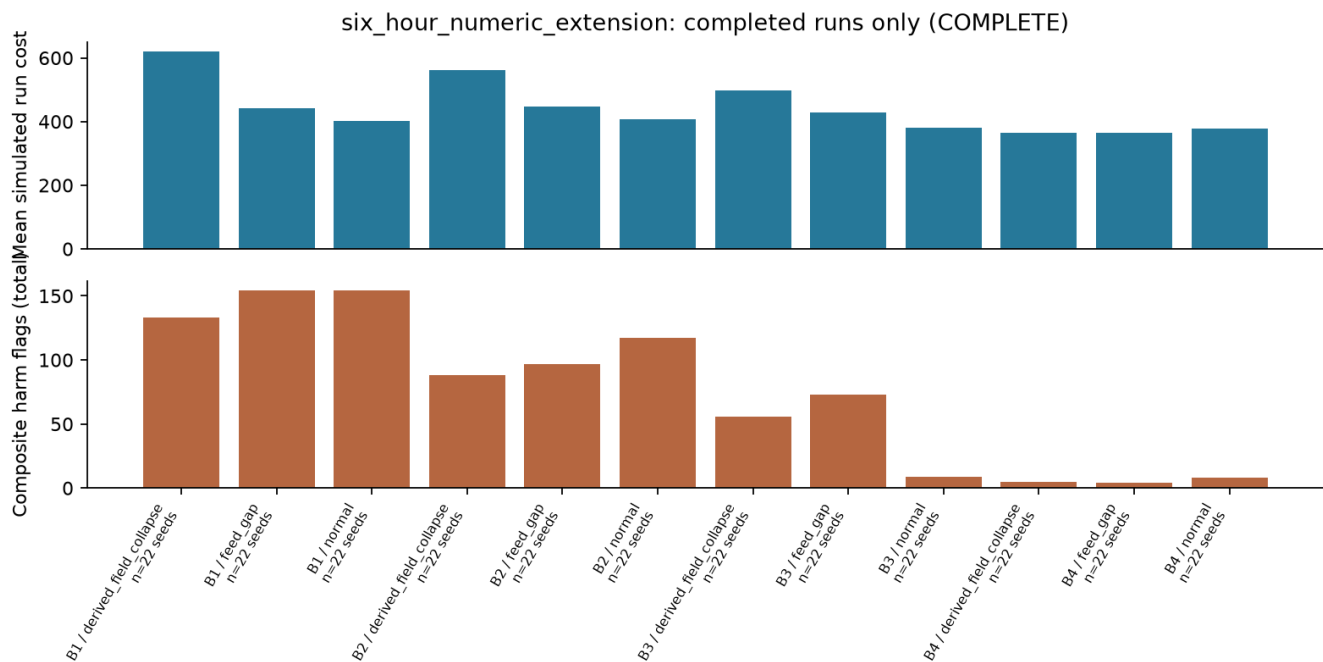
Policy / scenario	LLM calls	LLM errors	Fallback decisions
B1 / derived_field_collapse	0	0	0
B1 / feed_gap	0	0	0
B1 / normal	0	0	0
B2 / derived_field_collapse	0	0	0
B2 / feed_gap	0	0	0
B2 / normal	0	0	0
B3 / derived_field_collapse	0	0	0
B3 / feed_gap	0	0	0
B3 / normal	0	0	0
B4 / derived_field_collapse	0	0	0
B4 / feed_gap	0	0	0
B4 / normal	0	0	0

Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
B1 / derived_field_collapse	0.468	0.024	50.320	253.442	0.808
B1 / feed_gap	0.312	0.033	2.223	206.201	0.979
B1 / normal	0.377	0.029	1.891	205.175	0.993
B2 / derived_field_collapse	0.234	0.048	45.965	225.779	0.828
B2 / feed_gap	0.175	0.077	6.385	180.175	0.949
B2 / normal	0.175	0.066	1.904	183.753	0.976
B3 / derived_field_collapse	0.519	0.020	21.477	221.805	0.939
B3 / feed_gap	0.370	0.039	3.252	178.513	1.120
B3 / normal	0.494	0.024	3.342	175.851	1.180
B4 / derived_field_collapse	0.377	0.035	12.266	172.494	1.173
B4 / feed_gap	0.383	0.035	12.280	172.571	1.172

Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
B4 / normal	0.487	0.023	3.296	175.643	1.180

Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

The repository also wrote study_summary.json with replication-level cost/CVaR and utility. It is preserved in report_data.json. It may lag live worker results; this report's tables are refreshed from complete per-run artifacts. Run-level daily CVaR is not substituted for replication-level tail risk.



Descriptive simulated outcomes from completed cells; exposure and panel differ across stages. The harmful execution count combines distinct mechanisms detailed in the table.

six_hour_structured_controls — COMPLETE

Output: results/six_hour_structured_controls; config: results/run_configs/six_hour_structured_controls.yaml; dataset: data/processed/m5; series attested by completed-run manifest: 30. Seeds configured: 1; origins: 1; start index: 1830; warmup: 14; decision days: 2; approval: oracle with delay 1; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2016-01-18 at index1815 (d_1816); decisions 2016-02-02 to 2016-02-03 at indices1830-1831 (d_1831-d_1832). Source: data/processed/m5/calendar.csv.

Policy / scenario	n seeds / runs	Mean cost	Mean fill	Harm total	True violation / ref deviation	Holds total
B3 / derived_field_collapse	1 / 1	178.272	1.000	0	0 / 0	0
B3 / normal	1 / 1	97.066	1.000	0	0 / 0	0
B4 / derived_field_collapse	1 / 1	67.673	1.000	0	0 / 0	1
B4 / normal	1 / 1	96.499	1.000	0	0 / 0	0

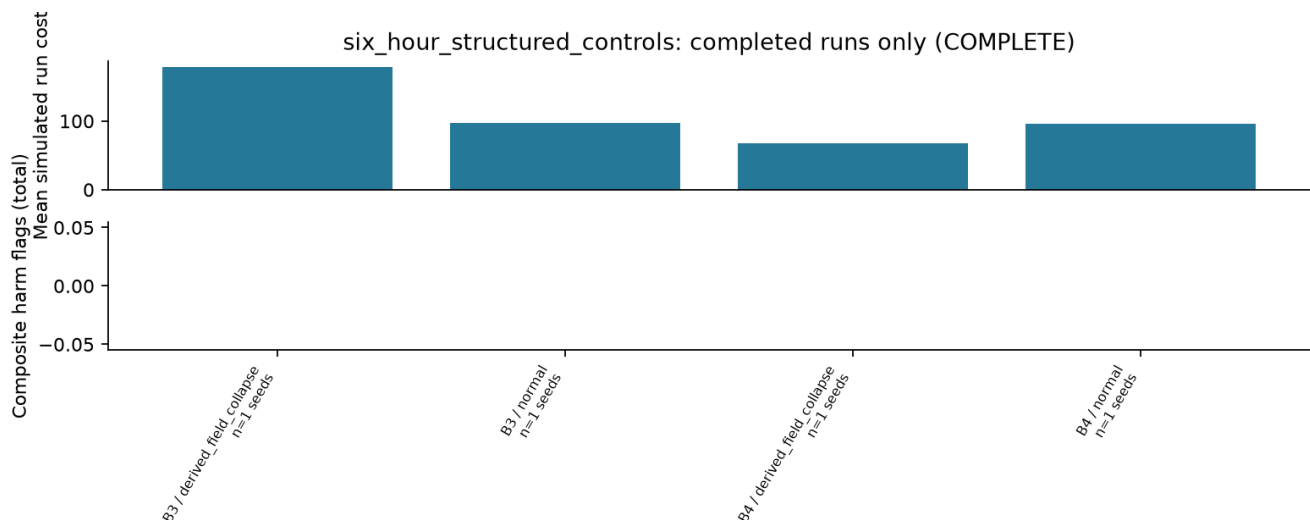
Policy / scenario	LLM calls	LLM errors	Fallback decisions
B3 / derived_field_collapse	0	0	0

Policy / scenario	LLM calls	LLM errors	Fallback decisions
B3 / normal	0	0	0
B4 / derived_field_collapse	0	0	0
B4 / normal	0	0	0

Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
B3 / derived_field_collapse	1.000	0.000	10.467	231.000	0.238
B3 / normal	1.000	0.000	1.526	224.000	0.246
B4 / derived_field_collapse	1.000	0.000	5.817	231.000	0.238
B4 / normal	1.000	0.000	1.526	225.000	0.244

Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

The repository also wrote study_summary.json with replication-level cost/CVaR and utility. It is preserved in report_data.json. It may lag live worker results; this report's tables are refreshed from complete per-run artifacts. Run-level daily CVaR is not substituted for replication-level tail risk.



Descriptive simulated outcomes from completed cells; exposure and panel differ across stages. The harmful execution count combines distinct mechanisms detailed in the table.

Paired inference — complete thirty-seed grids only

Inference remains pending for partial studies and is withheld for pilots with fewer than thirty configured independent seeds. For complete grids, repository paired_comparisons.csv uses per-seed origin averages, 5,000 paired bootstrap resamples (seed 42), unadjusted 95% intervals, Wilcoxon signed-rank p-values and Holm adjustment over the repository's within-study comparison family. These intervals do not establish statistical power or equivalence.

The frozen direct gate contrast is B10 minus B9 for cost, fill_rate and harmful_executions, across all scenarios configured in the local agent study. It uses the same paired bootstrap/Wilcoxon method after every configured seed and origin completes. Holm adjusts that contrast's fixed metric × scenario family separately from the repository's first-policy-reference family. Negative cost/harm differences favor B10; positive fill differences favor B10. Reported 95% bootstrap intervals are not adjusted for multiplicity. No significance is computed here from partial data.

main_study_stage1 paired comparisons

Pending: 20/480 runs complete. No paired significance table is generated for this incomplete grid.

foundation_30series paired comparisons

Pending: 0/360 runs complete. No paired significance table is generated for this incomplete grid.

llm_prose_reference_30series paired comparisons

Pending: 1/360 runs complete. No paired significance table is generated for this incomplete grid.

llm_prose_study_ollama paired comparisons

Pending: 0/540 runs complete. No paired significance table is generated for this incomplete grid.

llm_prose_study_reference paired comparisons

Repository reference comparisons (policy minus reference):

Scenario / contrast	Metric	Pairs	Mean difference	95% bootstrap CI	Holm p
derived_field_collapse / B4-B3	cost	30	-101.199	[-108.277, -94.681]	0.00000
derived_field_collapse / B4-B3	fill_rate	30	0.000	[0.000, 0.000]	1.00000
derived_field_collapse / B4-B3	harmful_executions	30	-0.833	[-1.067, -0.600]	0.00009
normal / B4-B3	cost	30	-1.190	[-3.139, 0.000]	0.71885
normal / B4-B3	fill_rate	30	0.000	[0.000, 0.000]	1.00000
normal / B4-B3	harmful_executions	30	0.000	[0.000, 0.000]	1.00000

llm_template_architectures_30series paired comparisons

Pending: 0/900 runs complete. No paired significance table is generated for this incomplete grid.

Validation and reproducibility checks

Recorded existing test-suite validation: passed=97, failed=0, skipped=1. Skip reason: optional anthropic package not installed. This is the recorded validation manifest; report generation does not rerun the suite.

Dataset acquisition provenance: scripts/fetch_m5.py downloaded the Nixtla public mirror and restored missing id/day labels without changing sales/prices/date values

Local Ollama recorded setup: {"binary_sha256": "c94aa4156b3d13e64ebc2efe5ea53f015384c882be776e6695cfb37fb180d5ad", "model_digest": "bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d", "model_identity": "results/run_configs/ollama_model_identity.json", "model_loaded_for_inference": false, "registry_connectivity": "Official manifest, model blob and other blobs downloaded over TLS; SHA256 and sizes verified; native pull succeeded", "version": "0.40.0"}

```
results/logs/ml_m5_replay.json: {"action_matches": true, "chain_valid": true, "decision_id":
"B4-normal-seed7-origin0:day1800", "held": false, "independent_violations": [], "mode": "cached external
evidence/tool replay; does not re-call the LLM or retrain forecasts", "objective_difference": 0.0,
"problem_reconstruction_matches": true, "problem_wire_hash_matches": true, "recorded_action_hash":
"cba7105ab172cb685fd367cb5b756f4a69a436786d8379d76bea884ec2644099", "replay_status": "exact action
match", "replayed_action_hash":
"cba7105ab172cb685fd367cb5b756f4a69a436786d8379d76bea884ec2644099", "state_certificate_matches":
true}
```

```
results/logs/ml_m5_trace_audit.json: {"artifacts_hash_verified": true, "completeness": 1.0,
"consumption_precision": 1.0, "consumption_recall": 0.9, "note": "Consumption is instrumentation evidence, not
proof that every consumed field was decisive. No neural chain-of-thought claims."}
```

Sample replay and trace-audit results validate their selected recorded decisions; they are not audits of every pending study run or re-executions of an LLM. Live functional and grounding validation are reported separately from these cached replay checks.

Reproducibility and artifact provenance

Current checkout commit: bbe3bbcad24291994fbc70fb329f165d2d97bb96. Tracked working-tree changes at report time: none. This attests the current checkout only; original per-run manifests do not necessarily contain a code SHA. Run configuration, environment and data evidence are taken from their actual artifacts. report_input_hashes.json records SHA-256 for every input file used by this report.

Launch protocol recorded code commit: bbe3bbcad24291994fbc70fb329f165d2d97bb96; model: ega-llama3.2:3b-cloud30; model revision: bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d. Models selected for the bounded study are identified separately in their current identity and frozen run configs. Findings from earlier local or historical hosted models cannot be inherited by a changed-model run.

Launch protocol's recorded scope: No model selection or gate retuning on evaluation windows Current required/auxiliary stage settings in the execution table take precedence if the registered scope has since expanded.

Launch protocol's recorded scope: Local 3b model is different from repository Cerebras study; cannot inherit historical provider findings Current required/auxiliary stage settings in the execution table take precedence if the registered scope has since expanded.

Launch protocol's recorded scope: M5 historical sales proxy; inventory, supply, documents and perturbations simulated Current required/auxiliary stage settings in the execution table take precedence if the registered scope has since expanded.

Launch protocol's recorded scope: Full 6120-run deterministic config and Chronos foundation baseline are separate extensions; not silently treated as executed Current required/auxiliary stage settings in the execution table take precedence if the registered scope has since expanded.

Raw M5 source	SHA-256 recorded by preparation manifest
calendar.csv	f9295292394975908e04ac77898c2304cbd499a2c466b2b9ce6f04079675bf16
sales_train_evaluation.csv	4b4a47c44c38380d2a9168216fea8c9ff2f31b1ddb772f8a0995952a038b8aa0
sell_prices.csv	5c7cf045b1abdbdc8d5007809c40b3682c5216075937f93e2cef01caddc4a30d9

The raw-source hashes above are recorded by the prepared-data manifests; report generation does not independently rehash large raw files. Per-panel catalog hashes and complete safe config snapshots are included in the HTML/Markdown and report_data.json. The input hash inventory permits detecting changes between report snapshots.

Lossless object packing and full-trace access

Current configured run artifacts with recognized packing manifests: 393. Packing stores the full original content-addressed JSON object inventory in artifacts/objects.zip and keeps the first three trace/certificate/proposal objects loose for standard reports. Missing loose copies of archived objects are not missing data when the inventory, archive and provenance verify. Summaries, daily rows, trace indices and the SQLite audit chain remain separate artifacts.

Stage	Packed/restored runs	Trace decisions available	Loose trace decisions	ZIP logical bytes
main_study_stage1	20	560	60	48,330,604
llm_prose_reference_30series	1	28	3	3,469,164
six_hour_agentic_prose	8	16	16	668,635
six_hour_numeric	96	672	296	28,882,510
six_hour_numeric_extension	264	1848	792	79,502,118
six_hour_structured_controls	4	8	8	357,870

The report checks current archive SHA/size, exact ZIP-member inventory, original inventory hash and unchanged run provenance. Archive SHA checks are reused only for the same file identity, size, modification time and expected checksum. Packing records attest byte-by-byte verification of original objects and the actual audit chain before removal. Full trace indices remain counted even when only three trace objects are loose.

Packing helper validation used a copied fixture, not experiment-output modification: original objects=367; retained after packing=9; archive bytes=2417610; actual outputs modified=False. These helper checks do not imply that all real study runs have been packed or independently audited.

This report's selected-object ZIP reader was checked on a copied fixture: archive/loose JSON equality=True; unknown reference refused=True; whole-run restoration performed by that check=False; actual experiment outputs modified=False.

Selected JSON can be read directly with this generator's PackedArtifactReader, which validates the requested object's original SHA-256 and byte length without whole-run restoration. Existing full replay, audit, grounding-artifact inspection and human-packet tools need restoration unless they explicitly use such a ZIP reader. Restore the CURRENT run directory with /workspace/MasterDissertation/.venv/bin/python /workspace/tools/m5_pack_artifacts.py --restore <CURRENT_RUN_DIRECTORY>; use the current merged location rather than a former worker path. This report performs no restoration or packing.

Stage	Input config SHA-256
m5_smoke	107d0cc30a834c4a04b45ef40b9ce9824924a27a8564bb740d6f2edf80b263e2
ml_m5_smoke	d64f153ca37d80bd161c42a02fd9693ee0568b674f7e6e409dd98c353a998bac
baseline_pilot	83955a0d49aa3c1ea79915d70ed22dee7fdbfcedf383cacae112815a95fbff7d
main_study_stage1	b7df6c95d2d9dbfd9498ff314410209aa990a6237d283a445cc76b535c9b7764
foundation_30series	a7b0542cbbe82b982bfa88b3365fc6382026a9140e7bbfad15d90f8104818a12
llm_architectures_ollama	3116ecb7b5859832f91fce050c01fb767121004700d20f1b9824bcb14fc20980
llm_prose_reference_30series	1cacb94fbc971705a142d945077b26be9b8c4780c446bcc7ab5b662fd3bbc0c7
llm_prose_study_ollama	0256bd45b8ec754c2fbf162f2aeaa0858fa3548cc8f5dbe0f0c77298529a6cfe
llm_prose_study_reference	0e57d47bcbcb5950eb9c821f3b6632e9d4e0fb5875acfd36176de88454765d943
llm_template_architectures_30series	cc0fc00f3f27c1ba4337c6449f10ae6780843e91e1f51b41010c3637baf7b14a
parallel_runner_validation	94a0c8ef793b031ed8e4b3ae872ffff6f304ce37c29a7ac9136f4e40197d8e1d

Stage	Input config SHA-256
six_hour_agentic_prose	4b70312c0f8aaec3bb29b185e612c356d0e18ddcaf07f3018ae76b738e2a0a78
six_hour_numeric	d56fedc68e3e4a1ebdf4fc8d2db576149ab28dbf5602a9affa49908ab647e91b
six_hour_numeric_exten sion	d400722b69a70caab08615840a5bea23019e3f9784dc243ef5c777897ceb2e88
six_hour_structured_cont rols	a3656550ba05e697ec0e99a5e084e93d0bce464b0d101f693d14f5302fb8634c

Refresh and completion criteria

Refresh this report after workers complete or readiness changes using `.venv/bin/python results/report/generate_report.py` from the repository root. The generator launches no experiments. Completed configured execution requires actual run artifacts and separately reviewed audit/error outcomes. A blocked or partial snapshot remains labeled as such; the six-hour subset does not complete unimplemented dissertation requirements.